

Advanced Java Programming

□ UNIT 2: Java Database Connectivity (JDBC)

□ University of Kigali

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UNIT 2. JAVA JDBC (Java Database Connectivity)

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2.1. Introduction to connecting Java to Database

- There are various software such NetBeans, Eclipse, BlueJ, Codenvy, etc. to code in Java and database software programs such as ADABAS, IBM DB2, Microsoft Access, Microsoft Excel.
- Microsoft SQL Server, MySQL, Oracle RDBMS, etc.
- **NetBeans/Eclipse and MySQL are one of the most popular.**
- **XAMPP** is a software distribution and an open-source cross platform web server which provides the Apache web server, MySQL database (actually MariaDB), Php and Perl (as command-line executables and Apache modules) all in one package. **It is available for Windows, MAC and Linux OSs. It is compiled and maintained by apache. The acronym XAMPP stands for;**

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2.1. Introduction to connecting JAVA to Database

- X – [cross platform operating systems] meaning it can run on any OS such as Mac OS , Windows , Linux etc.
- A – Apache - this is the web server software.
- M – MySQL - Database.
- P – PHP
- P – Perl – scripting language

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2.1. Introduction to connecting JAVA to Database

NetBeans is an integrated development environment (IDE) for Java. NetBeans allows applications to be developed from a set of modular software components called modules. **NetBeans runs on Windows, macOS, Linux and Solaris.** In addition to Java development, it has extensions for other languages like PHP, C, C++, HTML5, and JavaScript. Applications based on NetBeans, including the NetBeans IDE, can be extended by third party developers.

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2.2. How to install XAMPP on Windows

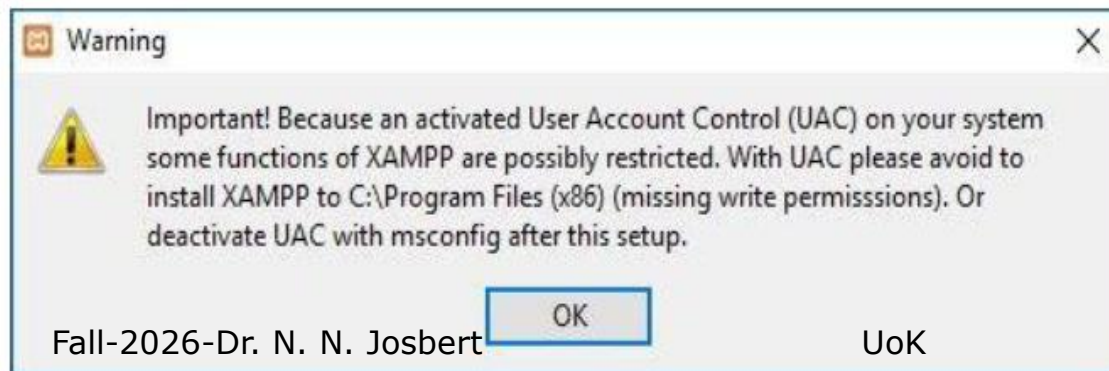
To download and install XAMPP on your Windows 10 computer, use the following steps:

1. Download XAMPP from **Apache Friends**.

<http://www.apachefriends.org/en/xampp-win-dows.html>

2. Double-click the file to run the installer.

3. Click the OK button on the warning to continue.



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2.2. How to install XAMPP on Windows

Click the **Next** button

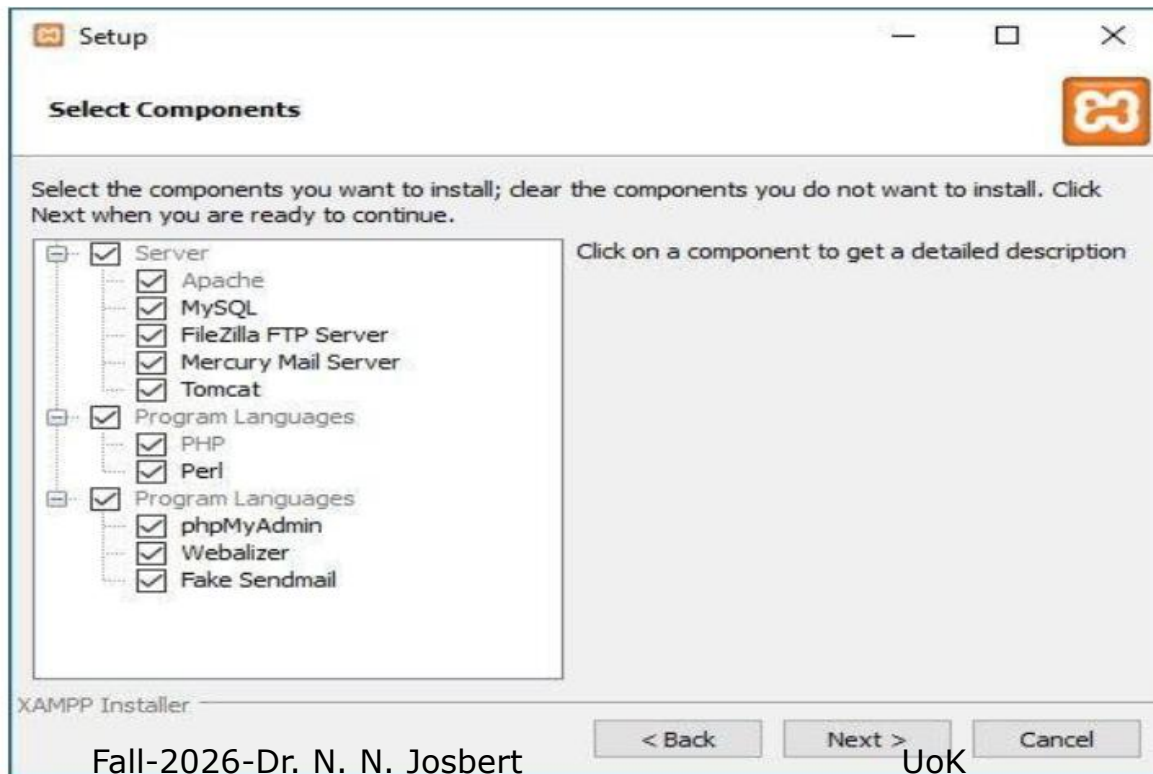


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2.2. How to install XAMPP on Windows

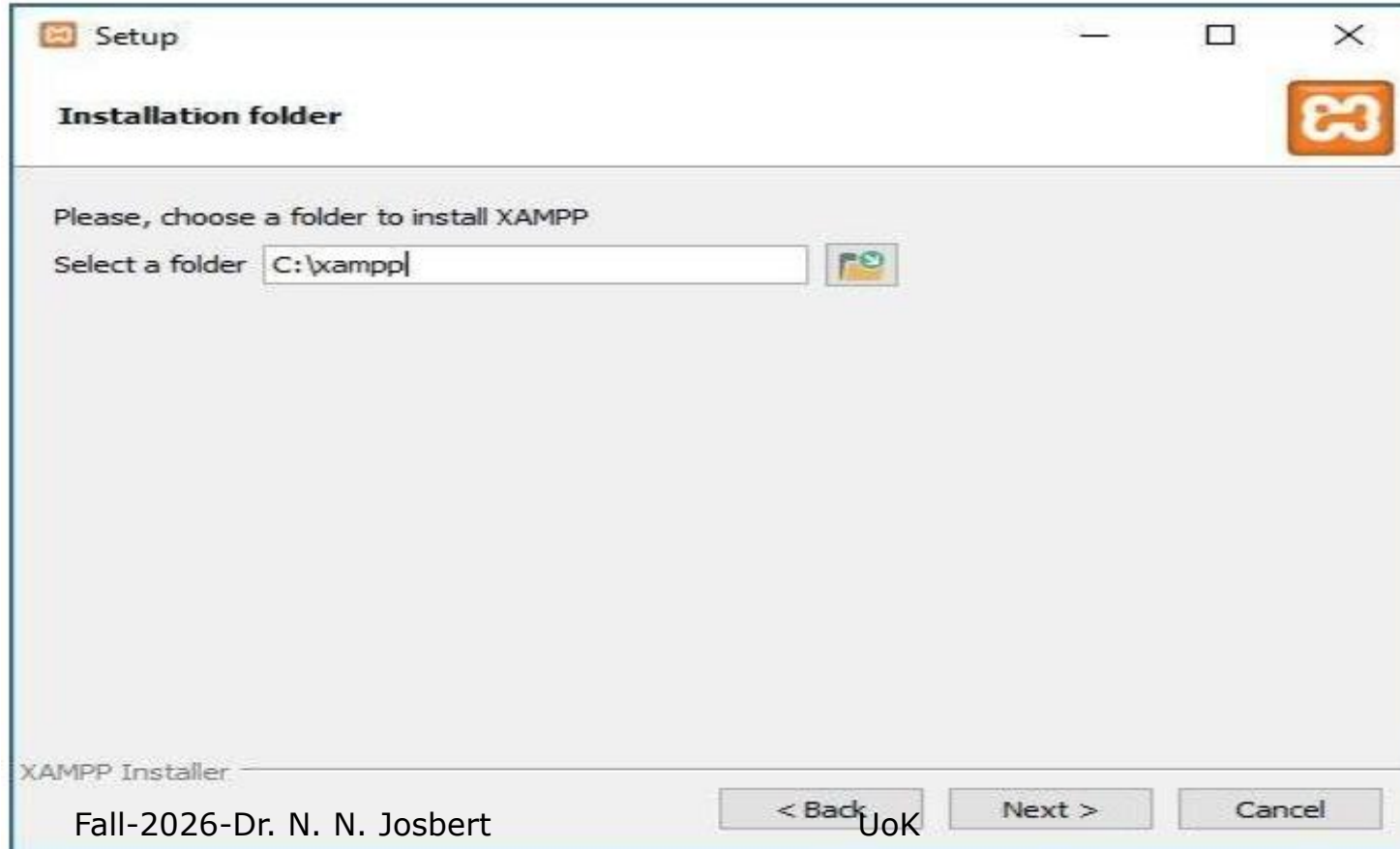
XAMPP offers a variety of components that you **can install, such as MySQL, phpMyAdmin, PHP, Apache, and more.** For the most part, you will be using most of these components, **as such it's recommended to leave the default options and click the Next button.**



UNIT 2. JAVA JDBC (Java Database Connectivity)

2.2. How to install XAMPP on Windows

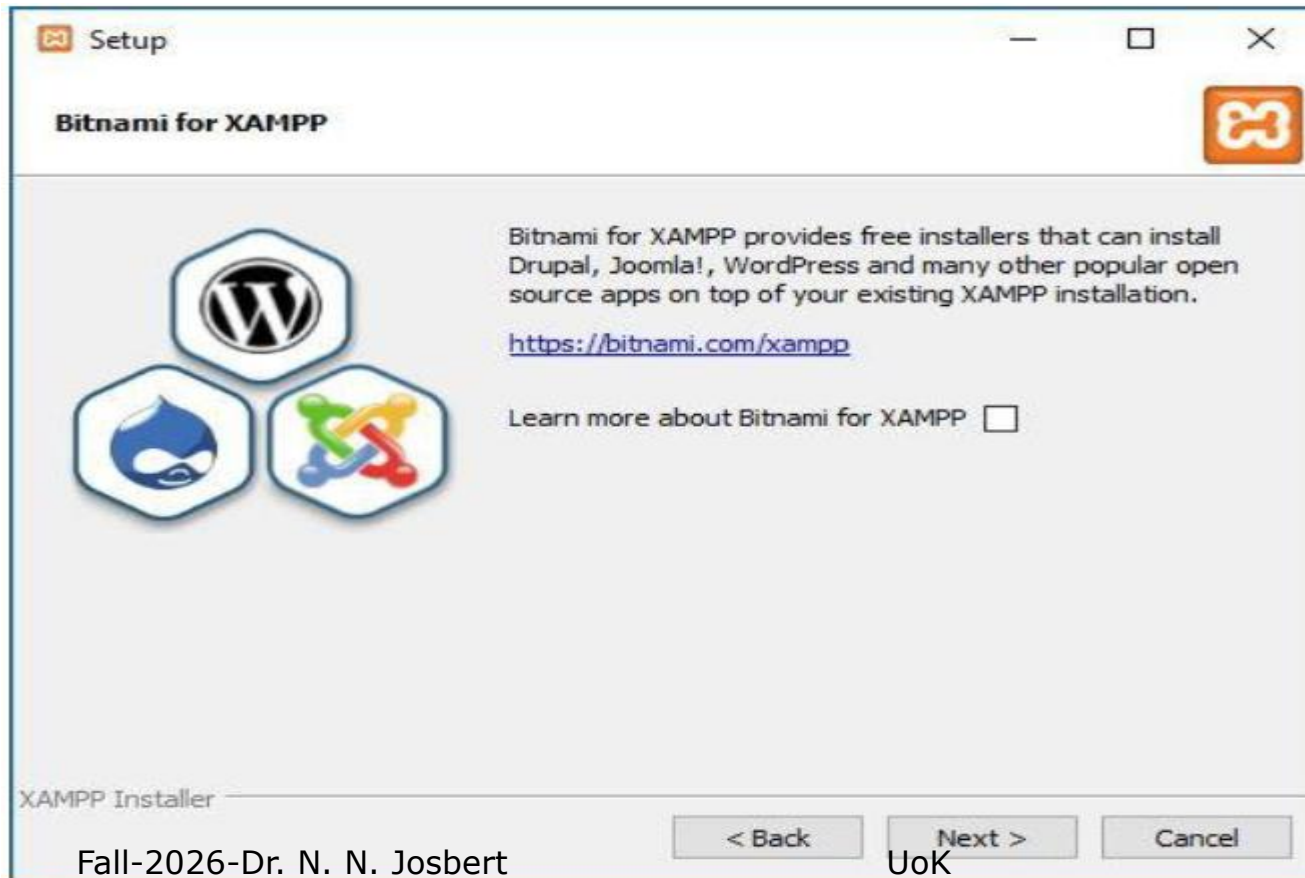
Use the default install location settings, or choose another folder to install the software, and click the **Next** button.



UNIT 2. JAVA JDBC (Java Database Connectivity)

2.2. How to install XAMPP on Windows

1. Clear the Learn more about Bitnami for XAMPP option.
2. Click the **Next** button.



UNIT 2. JAVA JDBC (Java Database Connectivity)

2.2. How to install XAMPP on Windows

Click the **Allow access** button to allow the app through the firewall (if applicable).



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2.2. How to install XAMPP on Windows

Click the **Finish** button.



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2.2. How to install XAMPP on Windows

1. Choose your language (English or German).
2. Click the **Save** button.



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2.2. How to install XAMPP on Windows

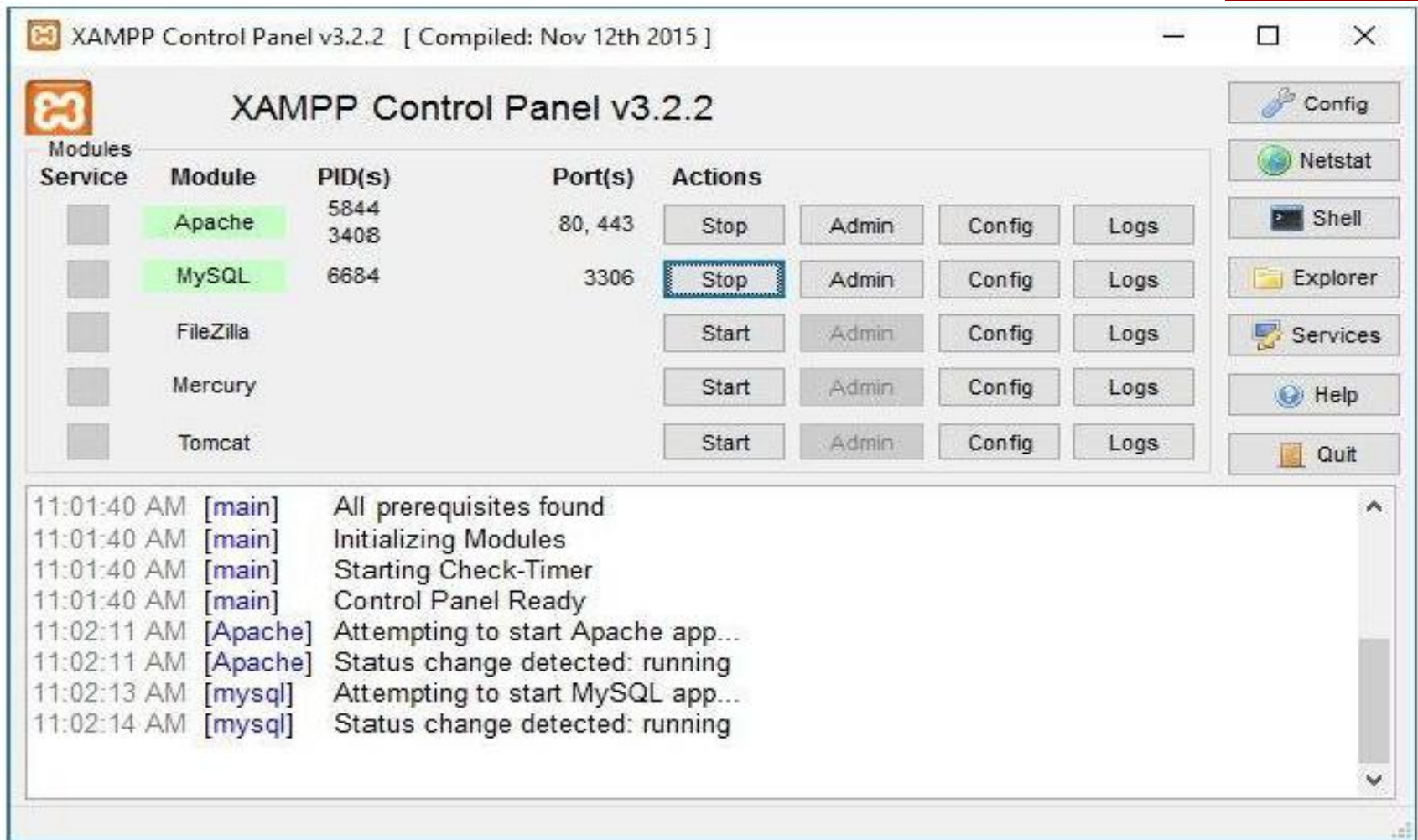
Once you've completed the steps, the XAMPP Control Panel will launch, and you can begin the web server environment configuration.

The XAMPP Control Panel includes three main sections. In **Modules**, you will find all the services available. **You can run each service by clicking the Start button.**

When you start some of the services, including Apache and MySQL, on the right side, you will also see the process ID number and TCP/IP port numbers each service is using. For example, by default **Apache uses TCP/IP port 80 and 443**, while **MySQL uses TCP/IP port 3306**.

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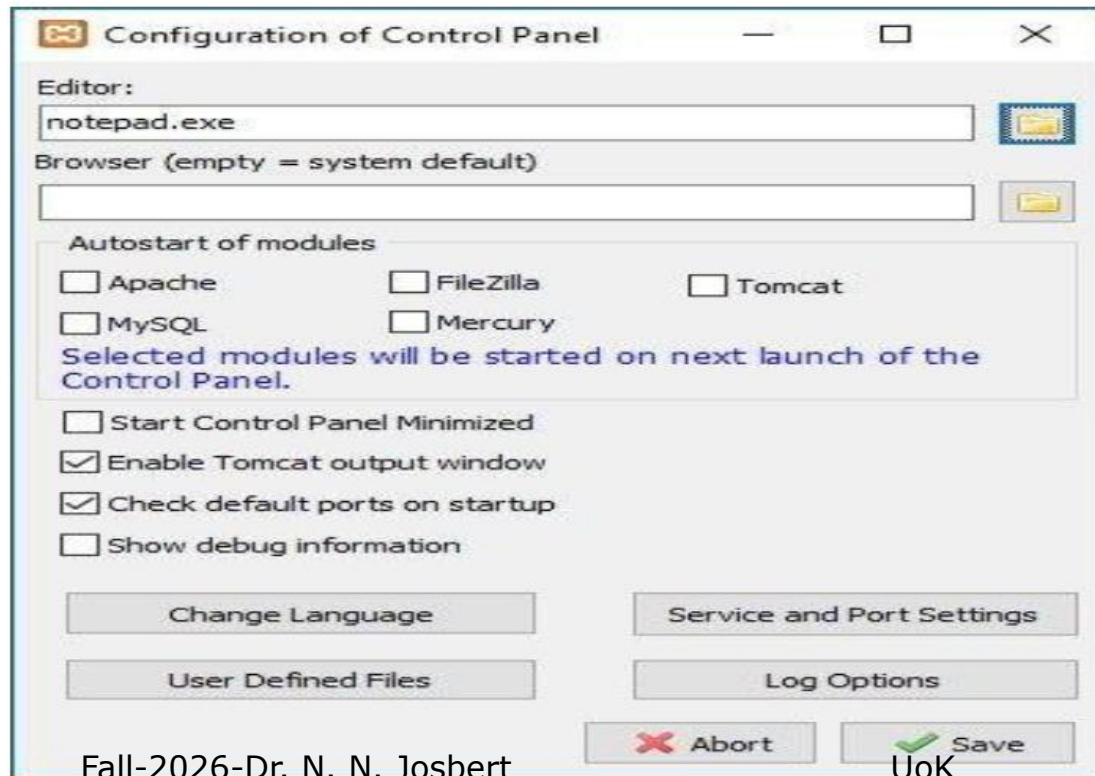
2.2. How to install XAMPP on Windows



UNIT 2. JAVA JDBC (Java Database Connectivity)

2.2. How to install XAMPP on Windows

In the right side there's a list of buttons to configure various aspects of the control panel, including **Config** to configure which modules will automatically want to start when you launch XAMPP.



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2.2. How to install XAMPP on Windows

Netstat will give you a list of services currently accessing the network, process ID and TCP/IP port information.

Netstat - TCP Listening sockets

Active socket New socket Old socket Refresh

Address	Port	PID	Name
0.0.0.0	80	10020	httpd.exe
0.0.0.0	135	744	svchost.exe
0.0.0.0	443	10020	httpd.exe
0.0.0.0	445	4	System
0.0.0.0	554	5856	wmpnetwk.exe
0.0.0.0	2869	4	System
0.0.0.0	3306	9888	mysqld.exe
0.0.0.0	5357	4	System
0.0.0.0	10243	4	System
0.0.0.0	49664	480	wininit.exe
0.0.0.0	49665	956	svchost.exe
0.0.0.0	49666	1076	svchost.exe
0.0.0.0	49667	1584	spoolsv.exe
0.0.0.0	49668	600	services.exe
0.0.0.0	49675	608	lsass.exe
10.1.1.3	139	4	System
10.1.1.3	52093	4236	SkypeHost.exe
10.1.1.7	139	4	System
10.1.1.7	14959	4236	SkypeHost.exe
10.1.1.7	51249	1492	explorer.exe
10.1.1.7	51253	6796	OneDrive.exe
10.1.1.7	52097	4236	SkypeHost.exe
10.1.1.7	52132	4312	svchost.exe
10.1.1.7	52155	6796	OneDrive.exe
10.1.1.7	52156	6796	OneDrive.exe
10.1.1.7	52157	7100	MicrosoftEdgeCP.exe
10.1.1.7	52158	7100	MicrosoftEdgeCP.exe
10.1.1.7	52159	7100	MicrosoftEdgeCP.exe

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2.2. How to install XAMPP on Windows

And there are also quick access buttons to open the shell command-line utility, XAMPP installation folder, services, and quit the app.

Lastly, you get the logs section, where you can get a glance of what happens every time you start a module or change a setting. **This is also the first place to look at when something isn't working.**



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2.2. How to install XAMPP on Windows

The default settings should work for most people using XAMPP to create a testing environment to run a website for personal use or a client. However, depending on your setup configuration, **you may need to change the TCP/IP port number for the Apache server, the database upload size, or set the password for phpMyAdmin.**

To change these settings, you'll need to use the **Config** button for the corresponding service. For example, you'll need to open the **httpd.conf** file to change the settings on the Apache server, and the **my.ini** file to change the settings for MySQL.

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2.3. How to start/stop MySQL

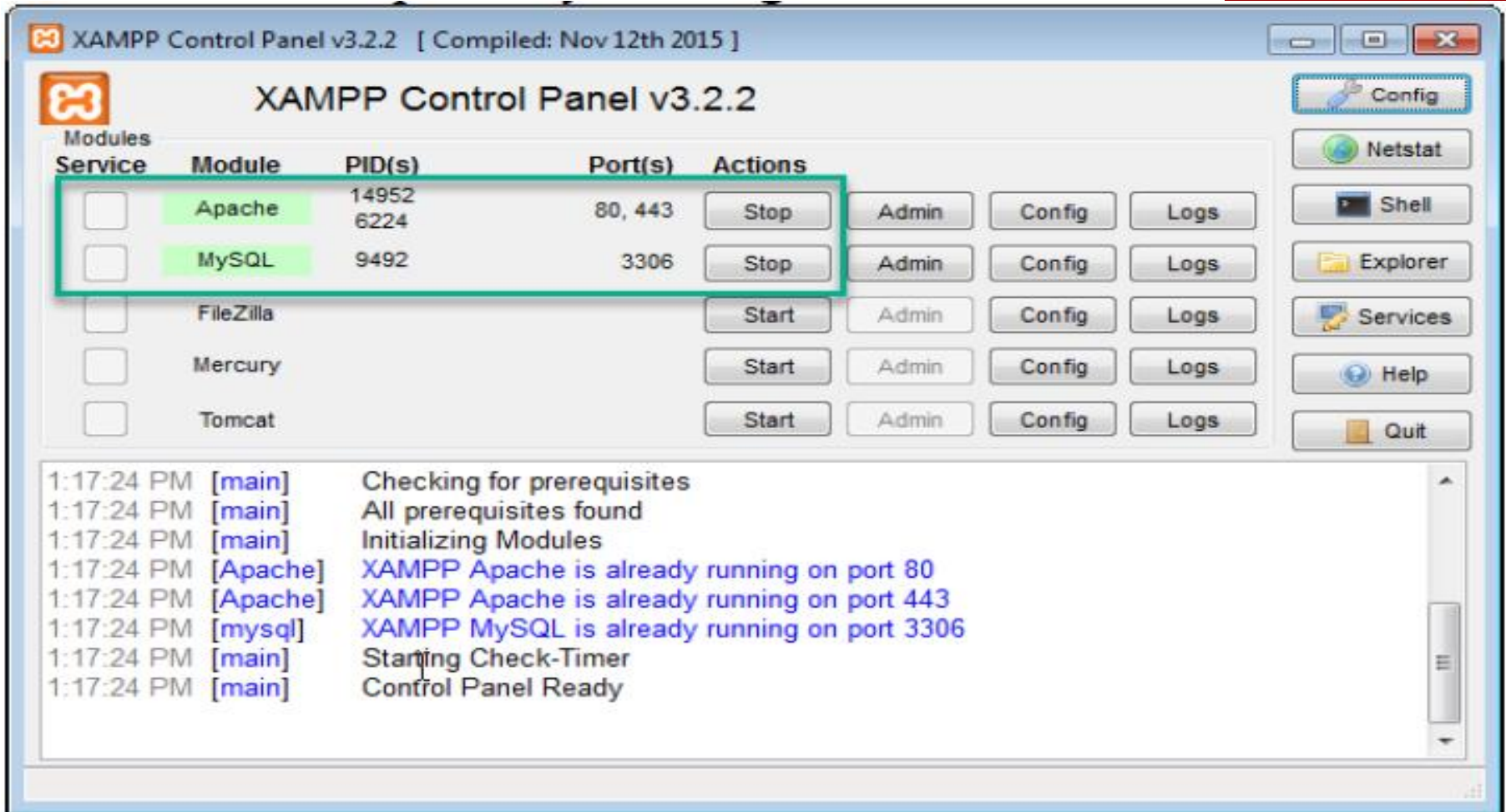
If you have MySQL already installed on your windows then go to services. msc file on your windows and right click the MySQL file and stop the service, now open your XAMPP and start MySQL. Now MySQL will start on the port 3306.

Or

To access phpMyAdmin from XAMPP you will need to make sure you have MySQL running in the XAMPP control panel by clicking the **start buttons under the Actions column:**

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2.3. How to start/stop MySQL

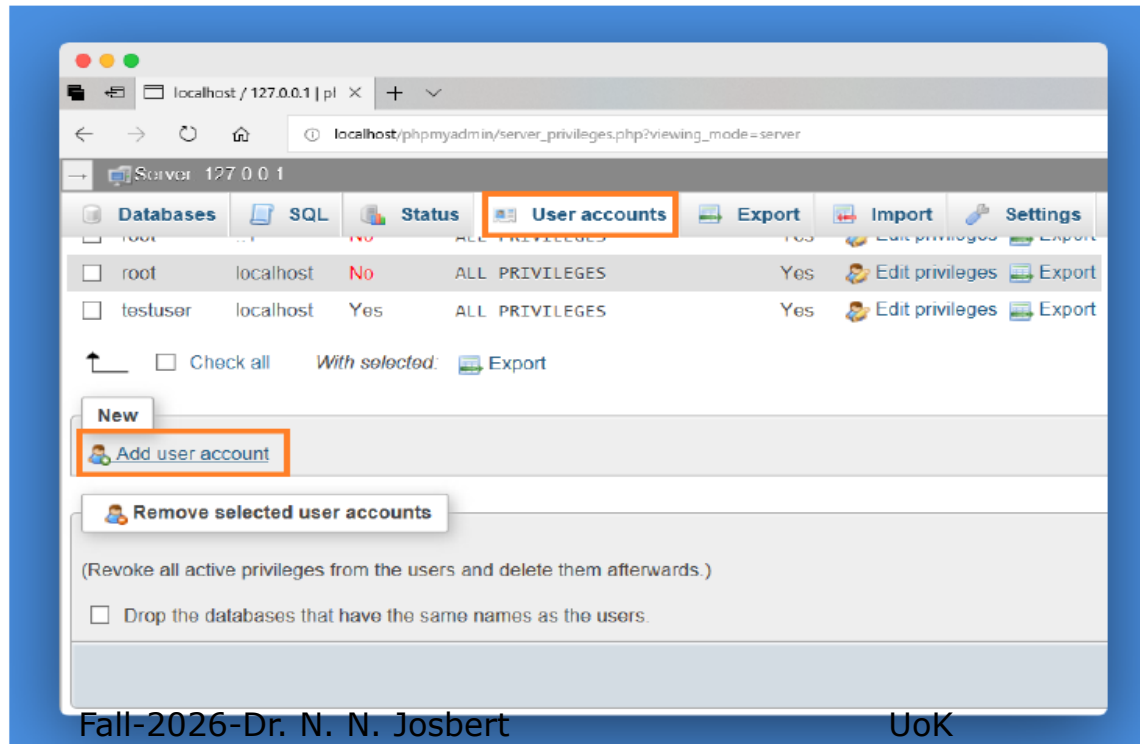


UNIT 2. JAVA JDBC (Java Database Connectivity)

2.4. Create User, Database and Table in MySQL

Create User:

To create a new database in MySQL, open browser, type localhost, then click on phpMyAdmin link, **click on users tab and create new user and database as shown in image below.**



UNIT 2. JAVA JDBC (Java Database Connectivity)

2.4. Create User, Database and Table in MySQL

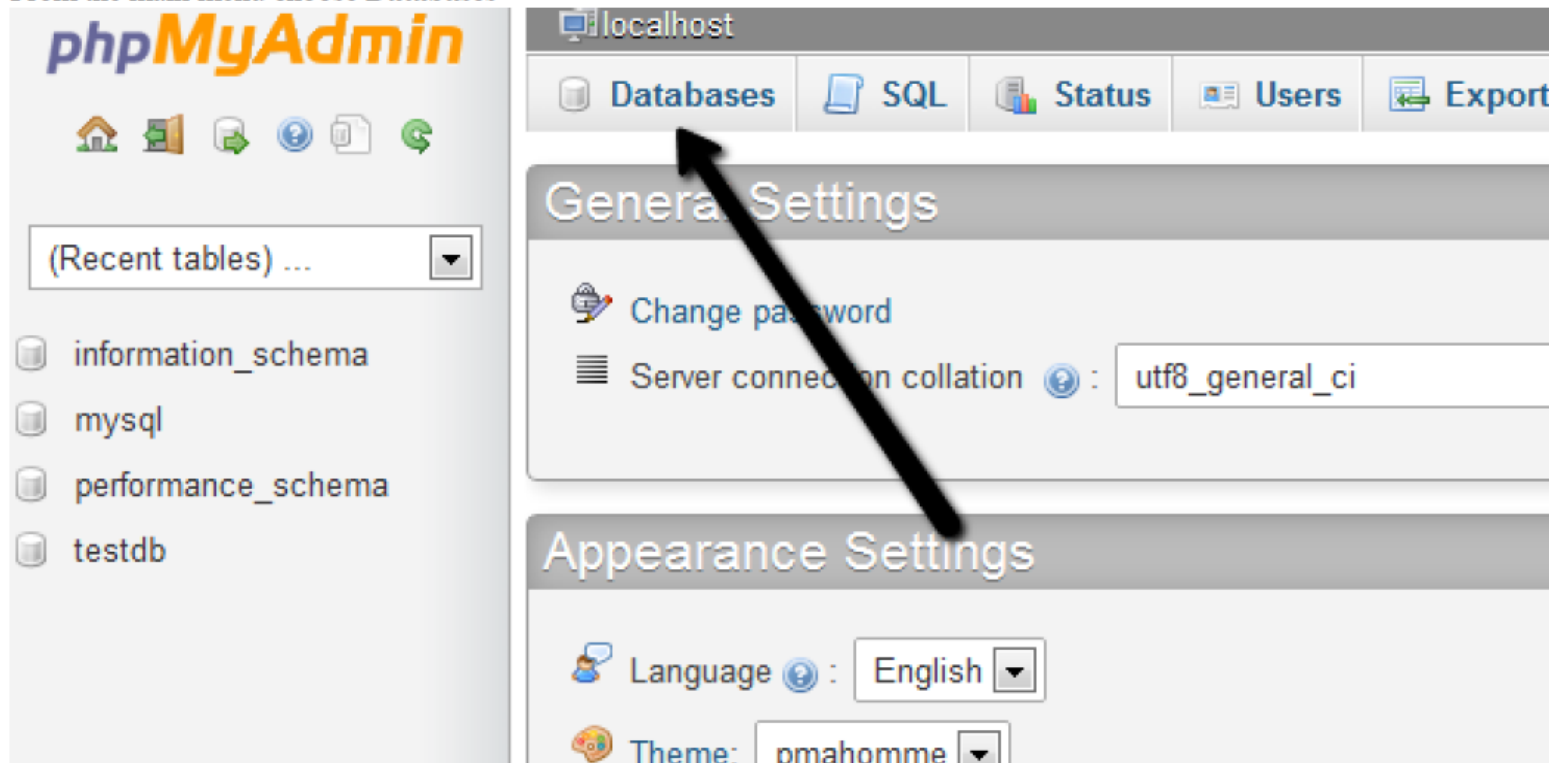
Click on Add new user and enter **username** for ex. **john**, **password** for ex. **john** & host-name **localhost**. the important step is select checkbox. Create database with same name and grant all privileges & global privileges, then click on go button.

UNIT 2. JAVA JDBC (Java Database Connectivity)

2.4. Create User, Database and Table in MySQL

To create database:

From the main menu choose **Databases**



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2.4. Create User, Database and Table in MySQL

and the create database field type in a name for your database for ex. **mydatabase**. Leave the collation drop down box if you wish to use the default MySQL schema collation. Click **Create**

Databases



Create database ?

mydatabase Collation Create

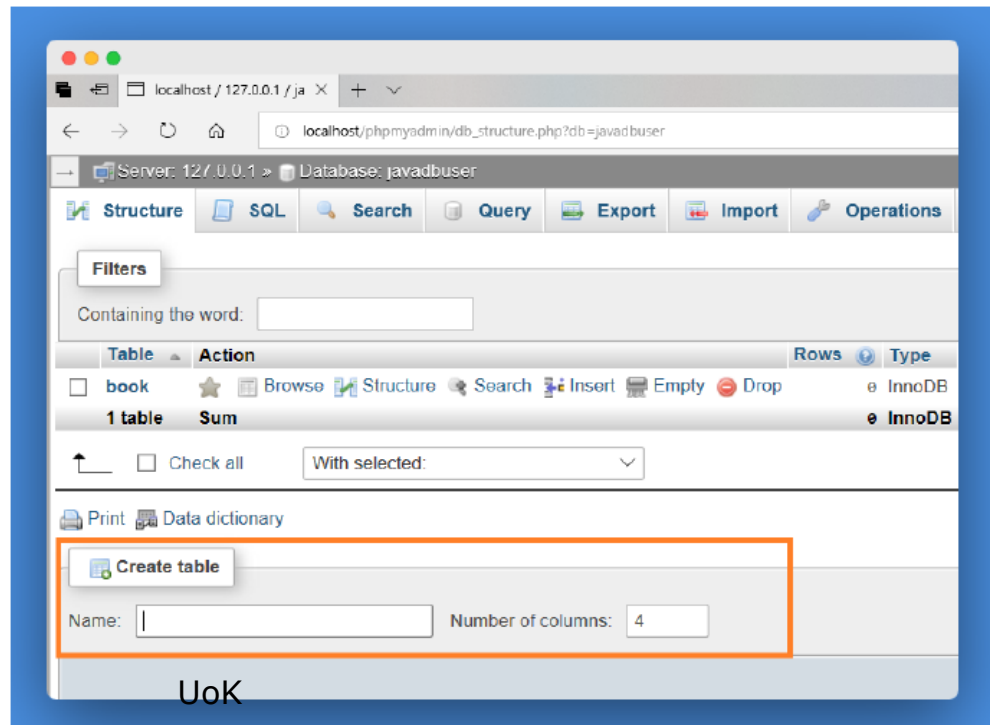
Your database will now be visible on the right-hand side under the list of available databases. To setup a new user login to access this database, click on **Users (here select john)** in the main menu. Choose the **Add User** option under the list of available server users.

UNIT 2. JAVA JDBC (Java Database Connectivity)

2.4. Create User, Database and Table in MySQL

To Create Table:

Click on Database Tab, click on database name which just created (i.e. **mydatabase**). then enter table **name (students)**, enter the number of fields/columns (i.e. **4**) then enter details of each field and click on **go button**.



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2.4. Create User, Database and Table in MySQL

Structure of persons table:

```
CREATE TABLE Persons (  
    ID int NOT NULL,  
    LastName varchar(255) NOT NULL,  
    FirstName varchar(255),  
    Age int,  
    PRIMARY KEY (ID)  
);
```

Structure of student table:

```
id INT(6) UNSIGNED AUTO INCREMENT PRIMARY KEY,  
firstname VARCHAR(30) NOT NULL,  
lastname VARCHAR(30) NOT NULL,  
email VARCHAR(50))
```

UNIT 2. JAVA JDBC (Java Database Connectivity)

2.5. How to Download & Install Java JDK 8 or 9 in Windows

Before proceeding with NetBeans install, you have to first install Java Development Kit(JDK) allows you to code and run Java programs.

Step 1) Go to link. Click on Download JDK. For java latest version.

Following are steps to install Java in Windows

UNIT 2. JAVA JDBC (Java Database Connectivity)

2.5. How to Download & Install Java JDK 8 or 9 in Windows

Java SE Downloads



Java Platform (JDK) 9



NetBeans with JDK 8

Java Platform, Standard Edition

Java SE 9.0.1
Java SE 9.0.1 includes important bug fixes. Oracle strongly recommends that all Java SE 9 users upgrade to this release.
[Learn more](#) ▶

- Installation Instructions
- Release Notes
- Oracle License
- Java SE Licensing Information User Manual
- Third Party Licenses
- Certified System Configurations
- Readme

JDK
DOWNLOAD ▼

Server JRE
DOWNLOAD ▼

JRE
DOWNLOAD ▼

Step 2) Next,

1. Accept License Agreement

2. Download latest Java JDK for your version (32 or 64 bit) of java for Windows.

UNIT 2. JAVA JDBC (Java Database Connectivity)

2.5. How to Download & Install Java JDK 8 or 9 in Windows

Java SE Development Kit 9 Downloads

Thank you for downloading this release of the Java™ Platform, Standard Edition Development Kit (JDK™). The JDK is a development environment for building applications, and components using the Java programming language.

The JDK includes tools useful for developing and testing programs written in the Java programming language and running on the Java platform.

See also:

- [Java Developer Newsletter](#): From your Oracle account, select **Subscriptions**, expand **Technology**, and subscribe to Java.
- [Java Developer Day](#) hands-on workshops (free) and other events
- [Java Magazine](#)

JDK 9.0.1 checksum

Java SE Development Kit 9.0.1

You must accept the [Oracle Binary Code License Agreement for Java SE](#) to download this software.

1 ☐ Accept License Agreement ☒ Decline License Agreement

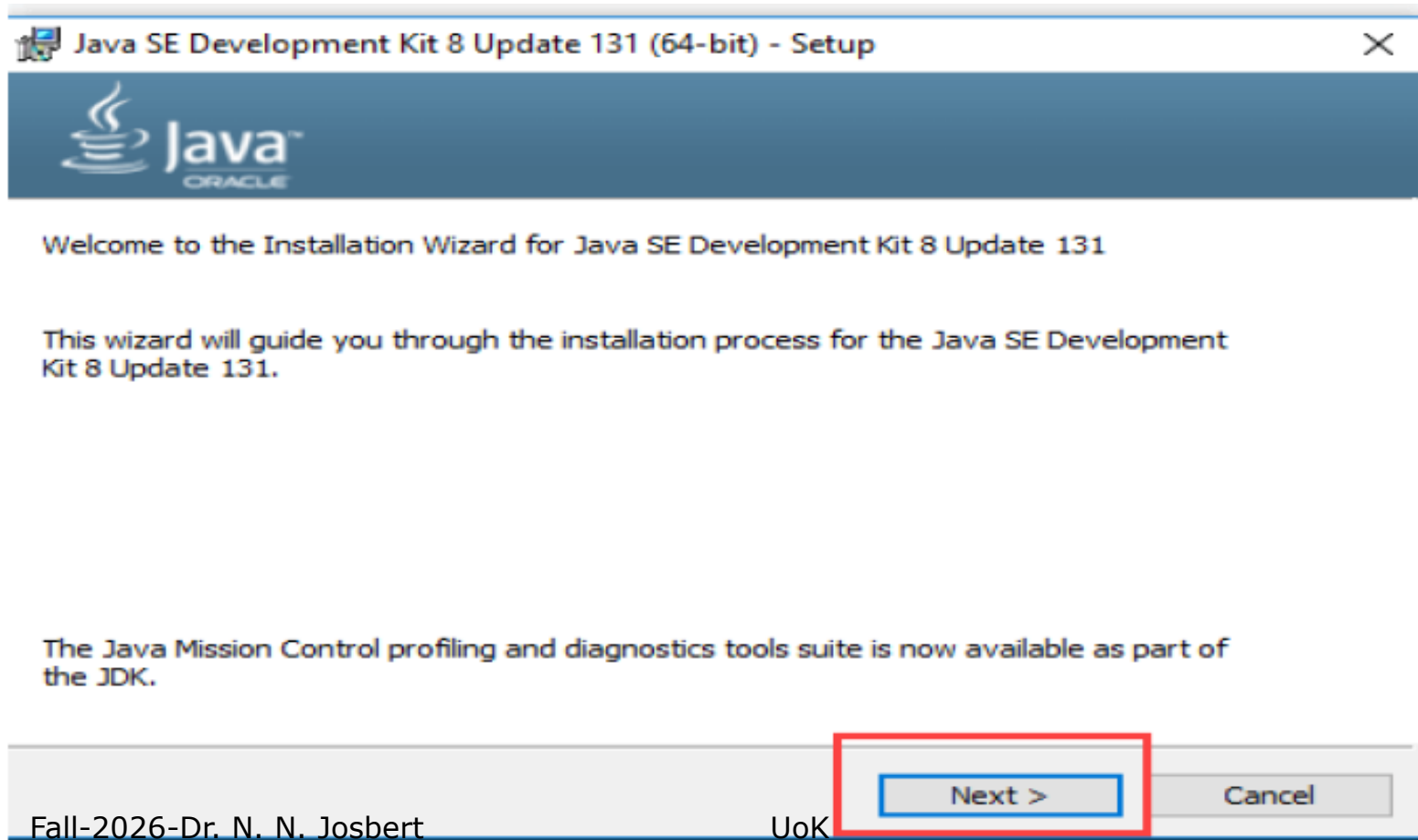
Product / File Description	File Size	Download
Linux	304.99 MB	jdk-9.0.1_linux-x64_bin.rpm
Linux	338.11 MB	jdk-9.0.1_linux-x64_bin.tar.gz
macOS	382.11 MB	jdk-9.0.1_osx-x64_bin.dmg
Windows	375.51 MB	jdk-9.0.1_windows-x64_bin.exe
Solaris SPARC	206.85 MB	jdk-9.0.1_solaris-sparcv9_bin.tar.gz

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UNIT 2. JAVA JDBC (Java Database Connectivity)

2.5. How to Download & Install Java JDK 8 or 9 in Windows

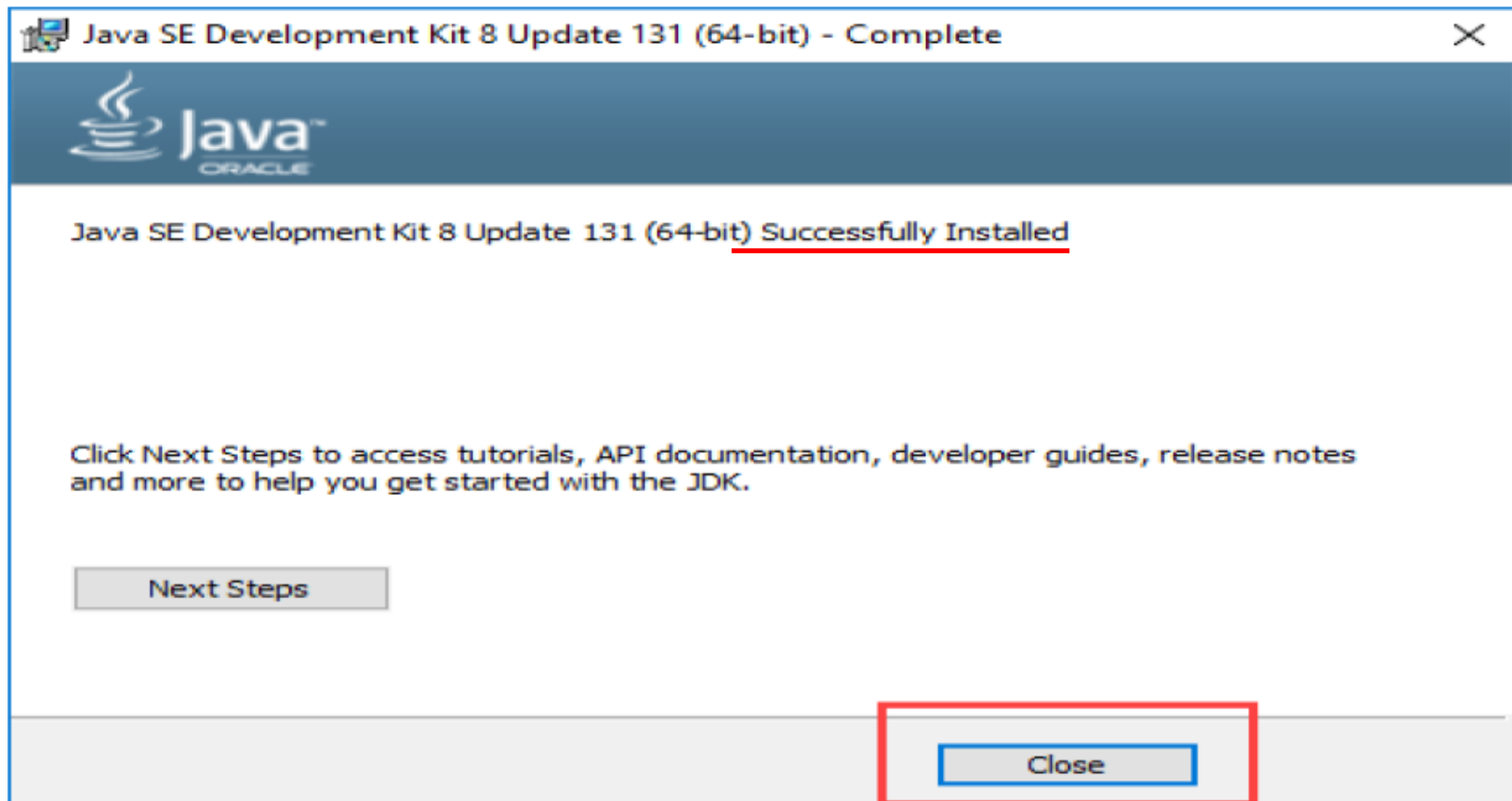
Step 3) Once the download is complete, run the exe for install JDK. Click Next



UNIT 2. JAVA JDBC (Java Database Connectivity)

2.5. How to Download & Install Java JDK 8 or 9 in Windows

Step 4) Once installation is complete click Close



UNIT 2. JAVA JDBC (Java Database Connectivity)

2.5. How to Download & Install NetBeans IDE

Step 1: First of all, download NetBeans IDE from official website of **NetBeans**: **DOWNLOAD IDE** When you will click on above given **DOWNLOAD** option so you will redirect on **NetBeans' official website**.



The screenshot shows the NetBeans IDE 8.2 Download page. The page has a navigation bar with links to NetBeans IDE, NetBeans Platform, Plugins, Docs & Support, Community, and Partners. Below the navigation bar, there is a search bar and a language selector. The main content area is titled 'NetBeans IDE 8.2 Download' and includes a table of supported technologies and download bundles. A red arrow points to the 'Download' button for the 'All' bundle.

Supported technologies *	Java SE	Java EE	HTML5/JavaScript	PHP	C/C++	All
NetBeans Platform SDK	•	•				•
Java SE	•	•				•
Java FX	•	•				•
Java EE		•				•
Java ME		•				•
HTML5/JavaScript		•	•	•		•
PHP			•	•		•
C/C++					•	•
Groovy						•
Java Card™ 3 Connected						•
Bundled servers						•
GlassFish Server Open Source Edition 4.1.1		•				•
Apache Tomcat 6.0.27		•				•

Download buttons: Download, Download, Download x86, Download x64, Download x86, Download x64, Download

Free, 95 MB Free, 197 MB Free, 108 - 112 MB Free, 108 - 112 MB Free, 107 - 110 MB Free, 221 MB

* You can add or remove packs later using the IDE's Plugin Manager (Tools | Plugins).

HTML5, PHP and C/C++ NetBeans bundles include Java Runtime Environment and do not require a separate Java installation.

JDK 9 is required for installing and running the Java SE, Java EE and All NetBeans Bundles. NetBeans 8.2 does not run on JDK9! You can download standalone JDK or download the latest JDK with NetBeans IDE Java SE bundle.

Using JDK 9 with NetBeans IDE? Go here.

Important Legal Information:

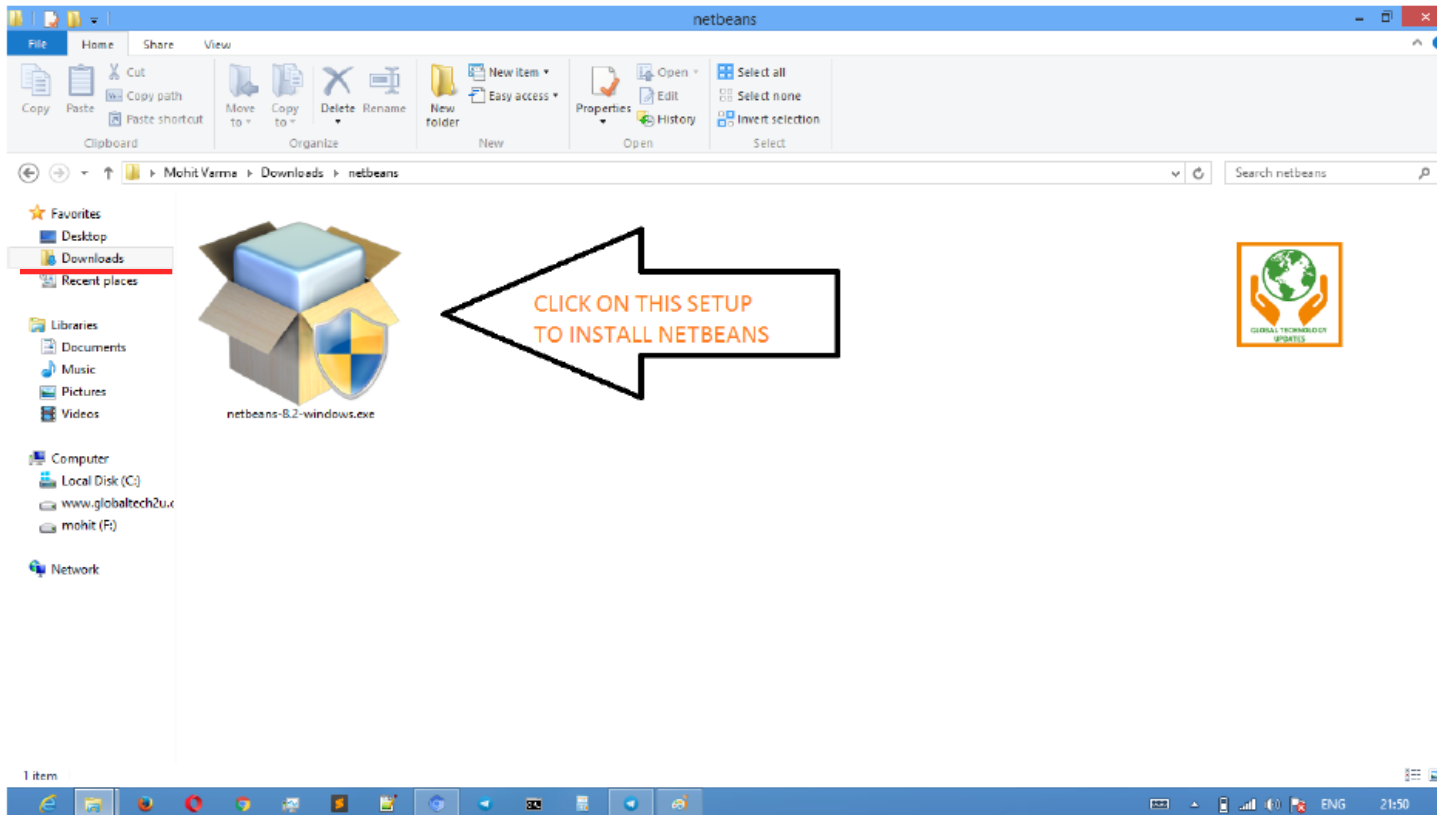
NetBeans Community Distributions are available under a Dual License consisting of the Common Development and Distribution License (CDDL) v1.0 and GNU General Public License (GPL) v2. Such distributions include additional components under separate licenses identified in the License file. See the Third Party License file for external components included in NetBeans and their associated licenses.

You can start developing applications based on the NetBeans Platform using the NetBeans IDE for Java SE. Learn more about the NetBeans Platform. NetBeans source code and binary builds without bundled runtimes are also available in zip file format. See also instructions on how to build the IDE from sources or installation instructions.

UNIT 2. JAVA JDBC (Java Database Connectivity)

2.5. How to Download & Install NetBeans IDE

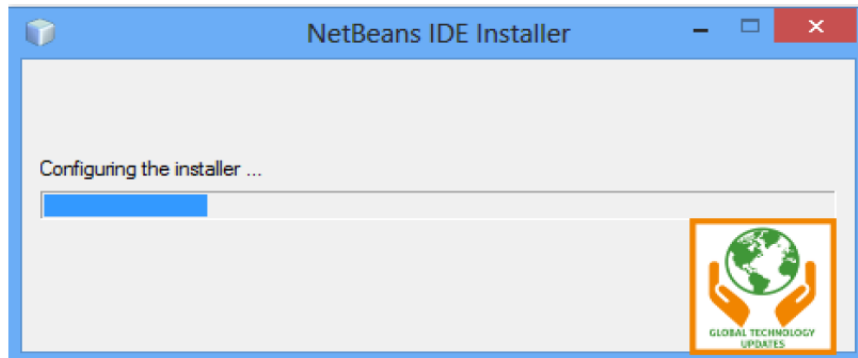
Step 2: Go at **your download location** and there you will see .exe file, double click on setup to further action.



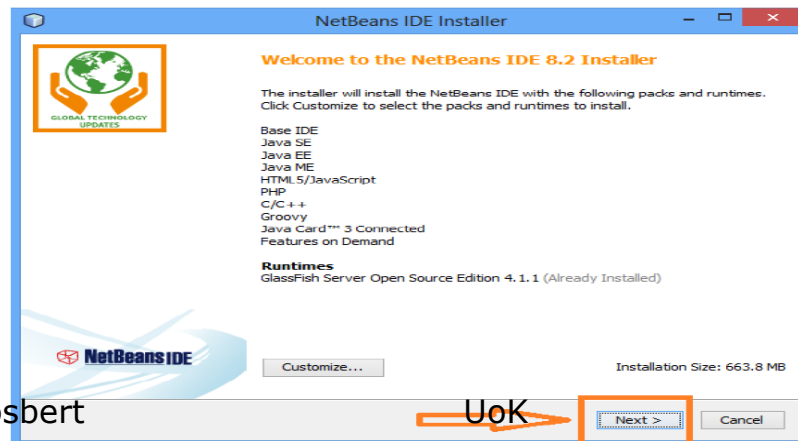
UNIT 2. JAVA JDBC (Java Database Connectivity)

2.5. How to Download & Install NetBeans IDE

It will take little moment to configuration of NetBeans IDE installer.



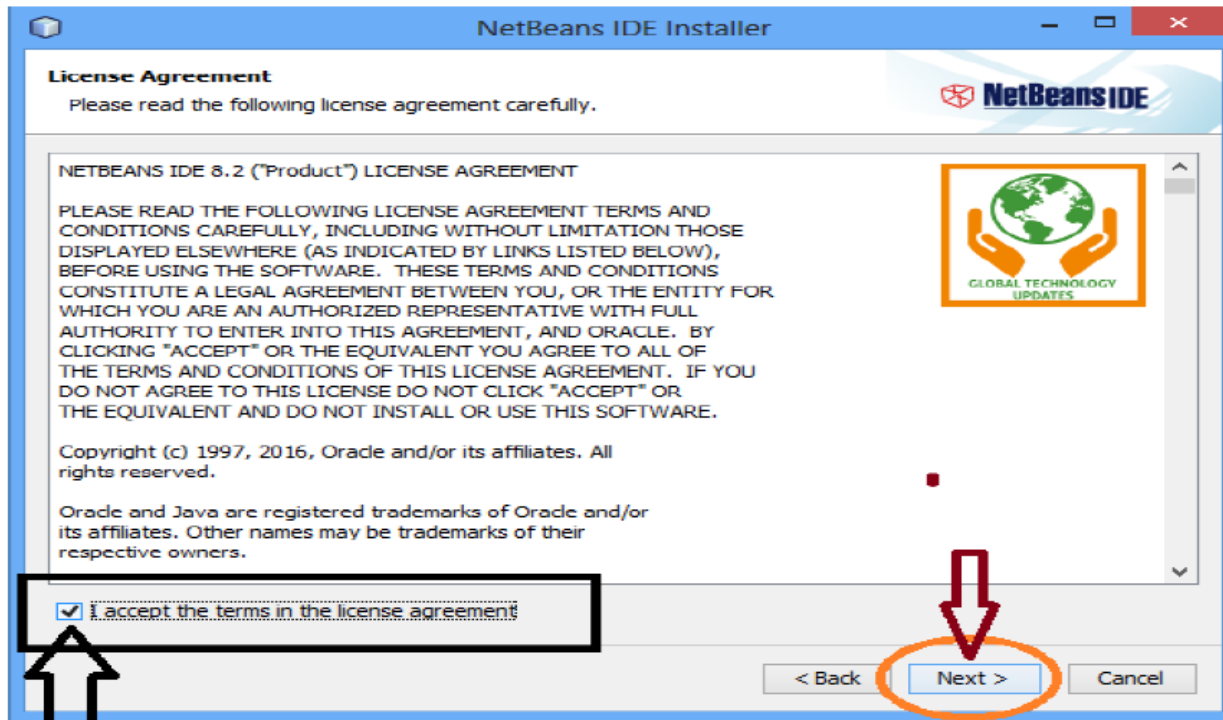
Step 3: Finally loaded NetBeans IDE installer at your system, click on “next”



UNIT 2. JAVA JDBC (Java Database Connectivity)

2.5. How to Download & Install NetBeans IDE

Step 4: Here is simple “License Agreement”, **accept License Agreement and click on “Next”**

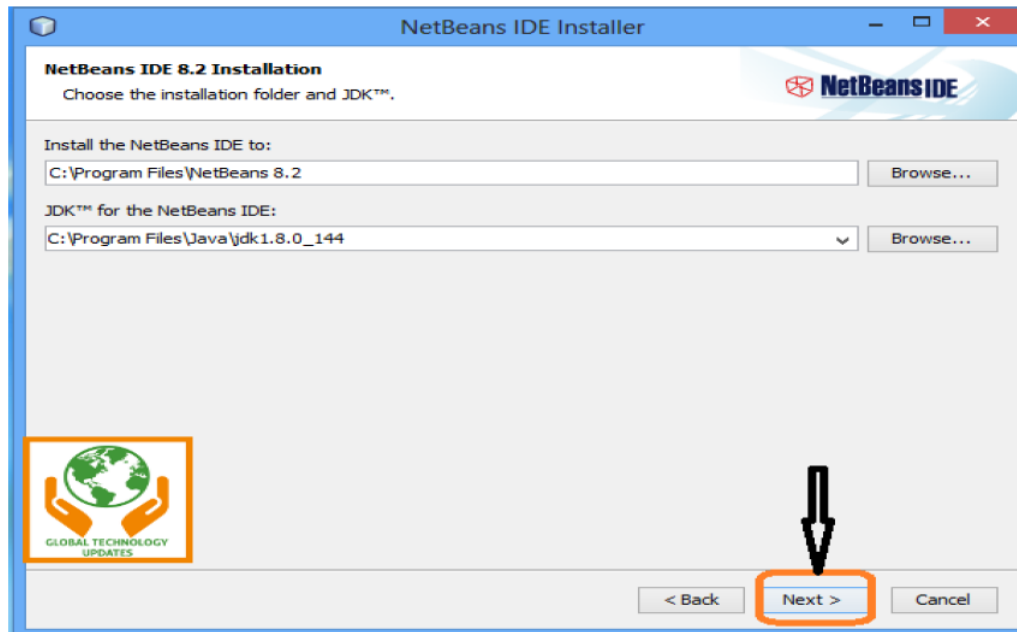


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2.5. How to Download & Install NetBeans IDE

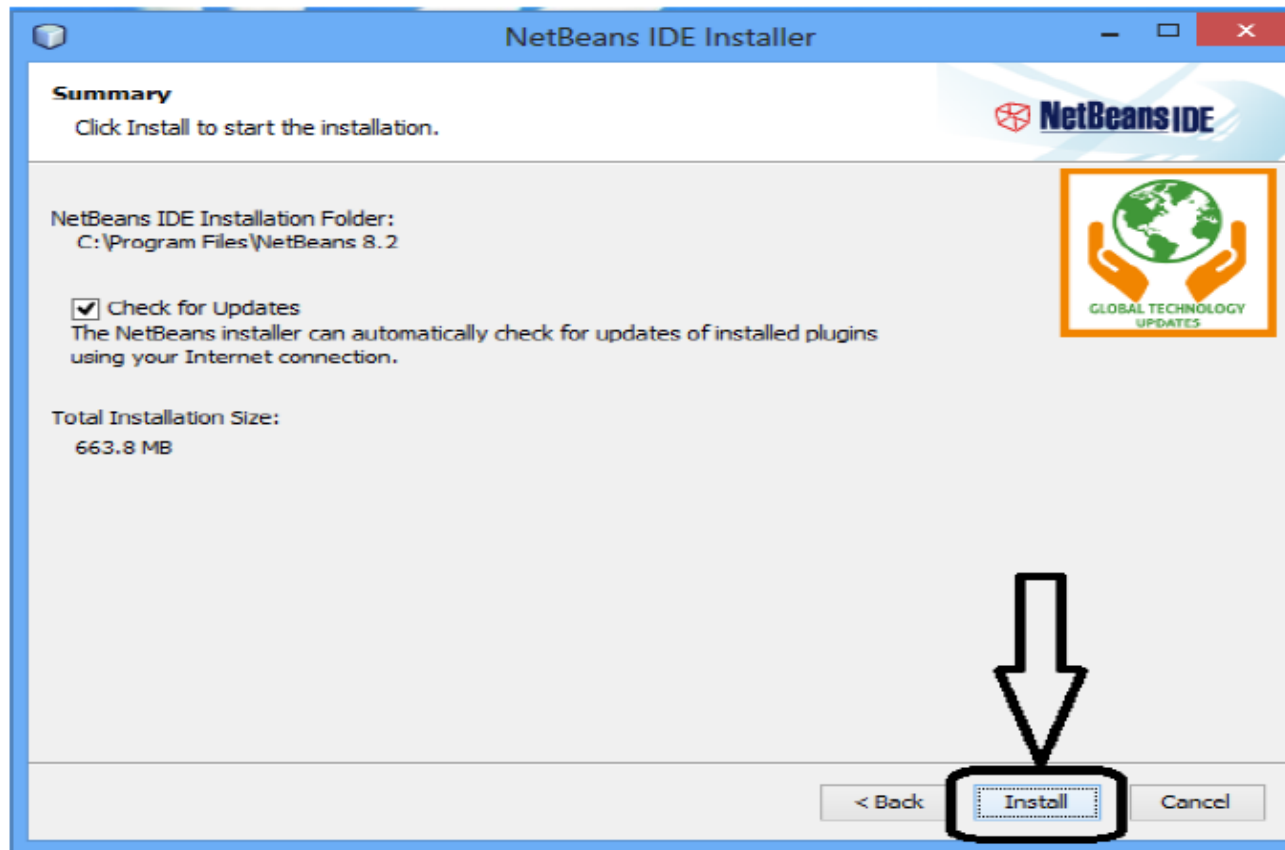
Step 5: Leave “Browse ...” Default and click on “**Next**”. If you want to install IDE in another directory so click on browse and define the place where you want and this process is same for JDK for NetBeans IDE.



UNIT 2. JAVA JDBC (Java Database Connectivity)

2.5. How to Download & Install NetBeans IDE

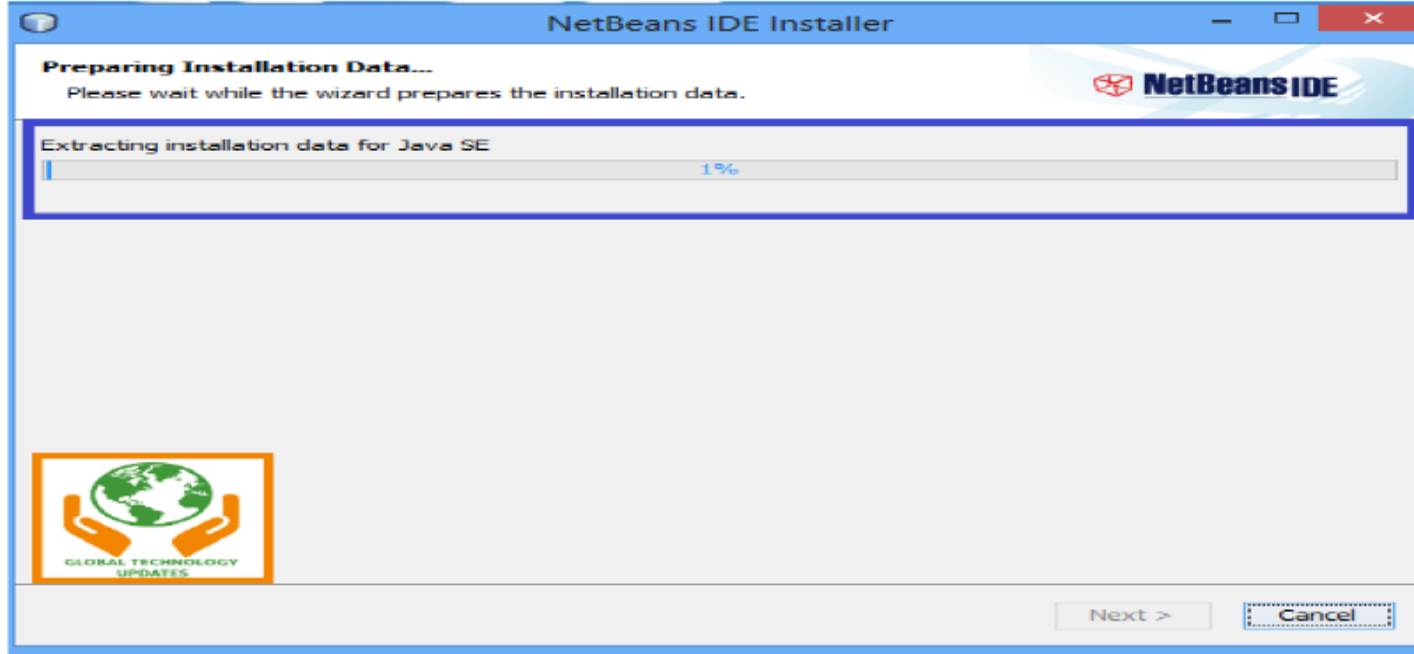
Step 6: Click on “**Install**” NetBeans installation process will start within a second.



UNIT 2. JAVA JDBC (Java Database Connectivity)

2.5. How to Download & Install NetBeans IDE

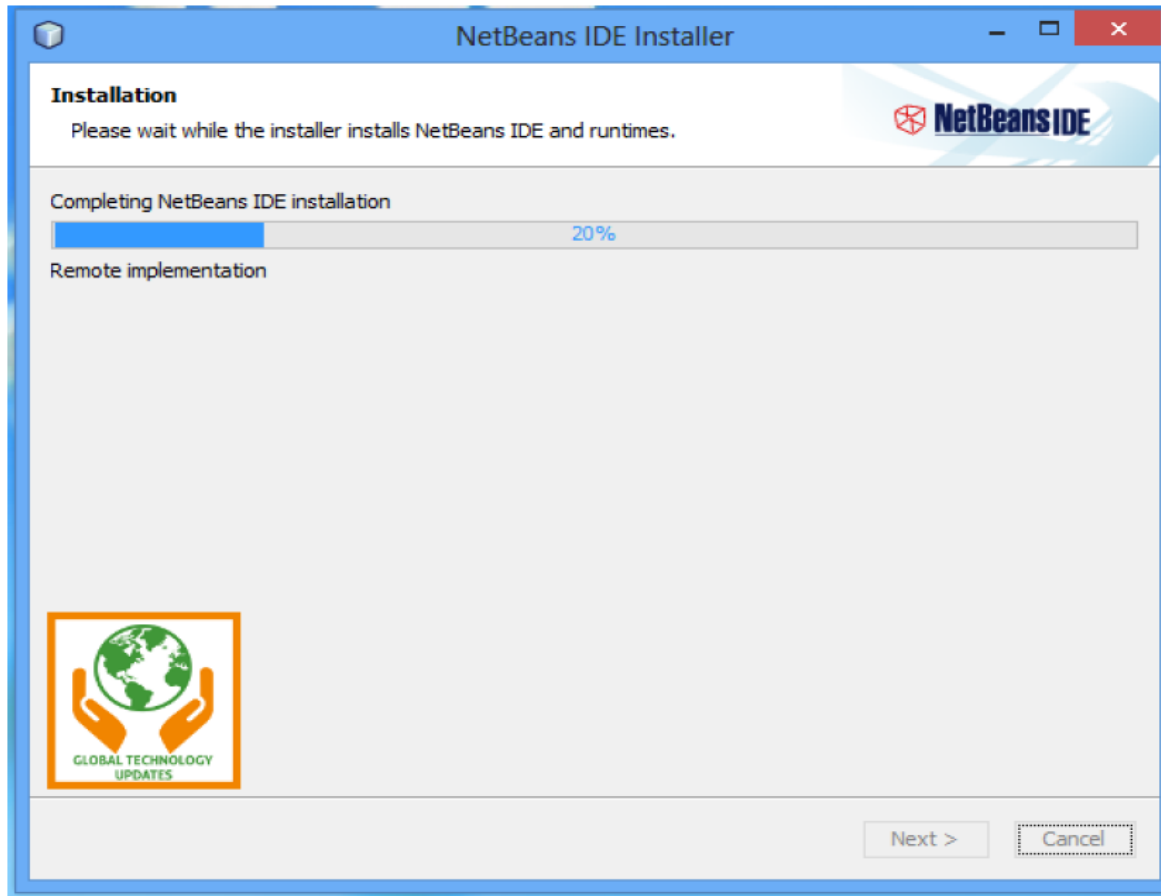
You can see here installation process started and this process takes some moment to complete, installation process is completed only 1%.



UNIT 2. JAVA JDBC (Java Database Connectivity)

2.5. How to Download & Install NetBeans IDE

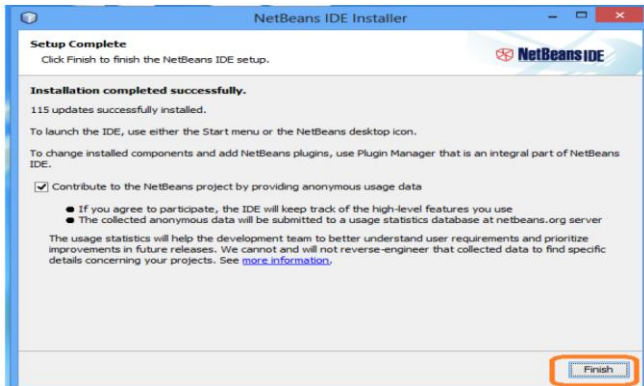
Installation process completed 20%.



UNIT 2. JAVA JDBC (Java Database Connectivity)

2.5. How to Download & Install NetBeans IDE

And installation completed successfully, now click on “Finish” for further action.



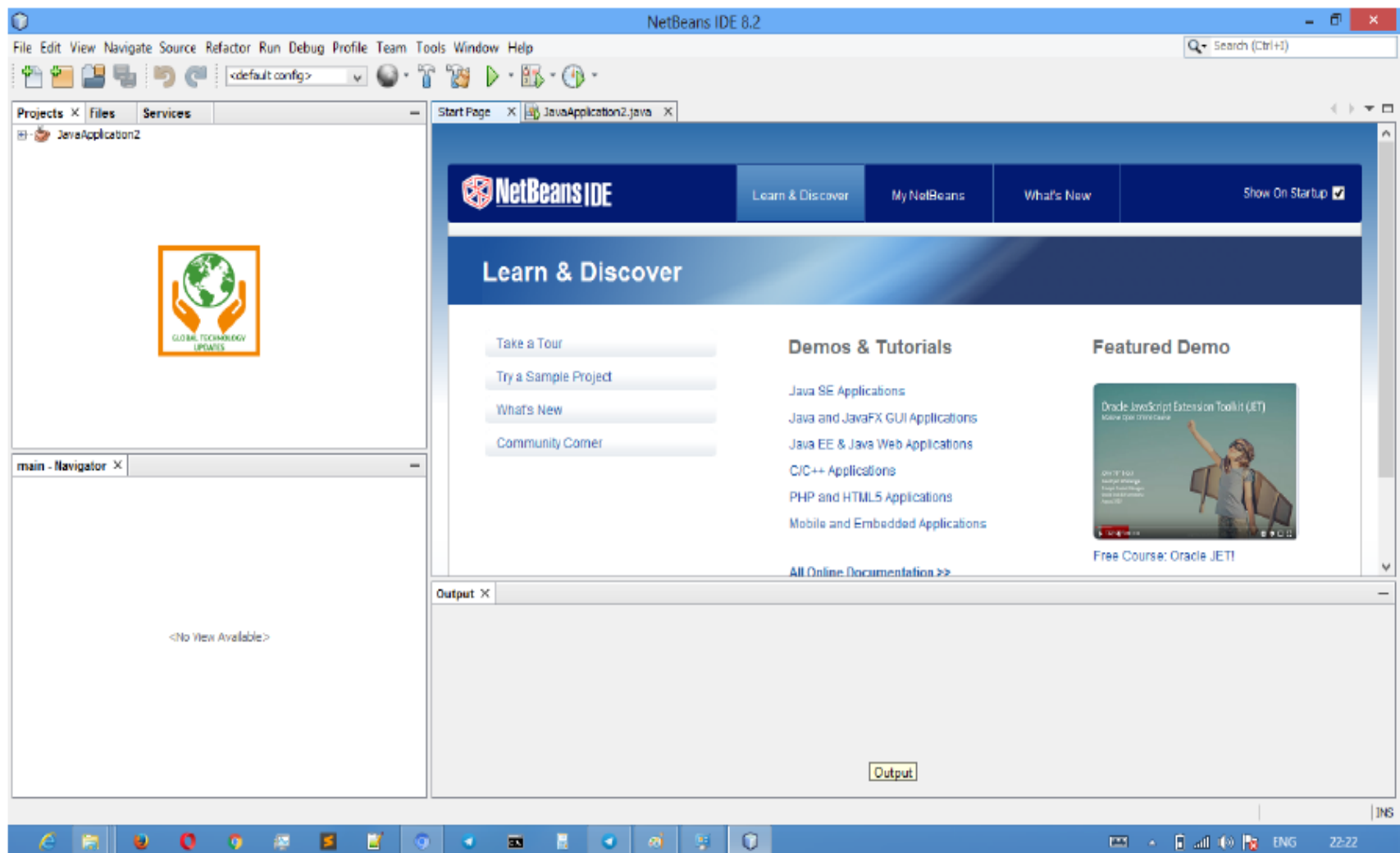
Now NetBeans IDE will start automatically on clicking “Finish” as you can see in below figure.



This is first interface of NetBeans IDE as you can see.

UNIT 2. JAVA JDBC (Java Database Connectivity)

2.5. How to Download & Install NetBeans IDE

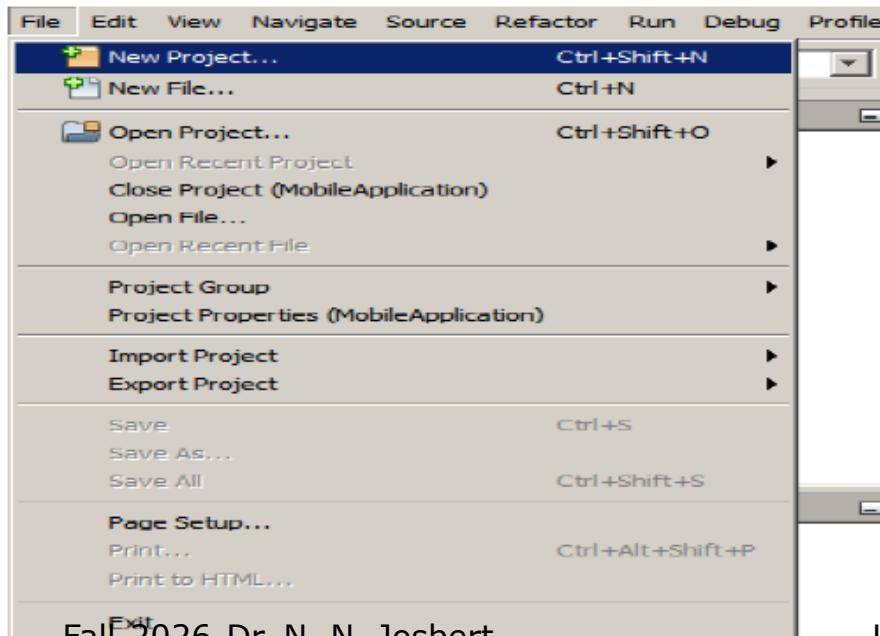


UNIT 2. JAVA JDBC (Java Database Connectivity)

2.6. Creating Java project in NetBeans

To create an IDE project:

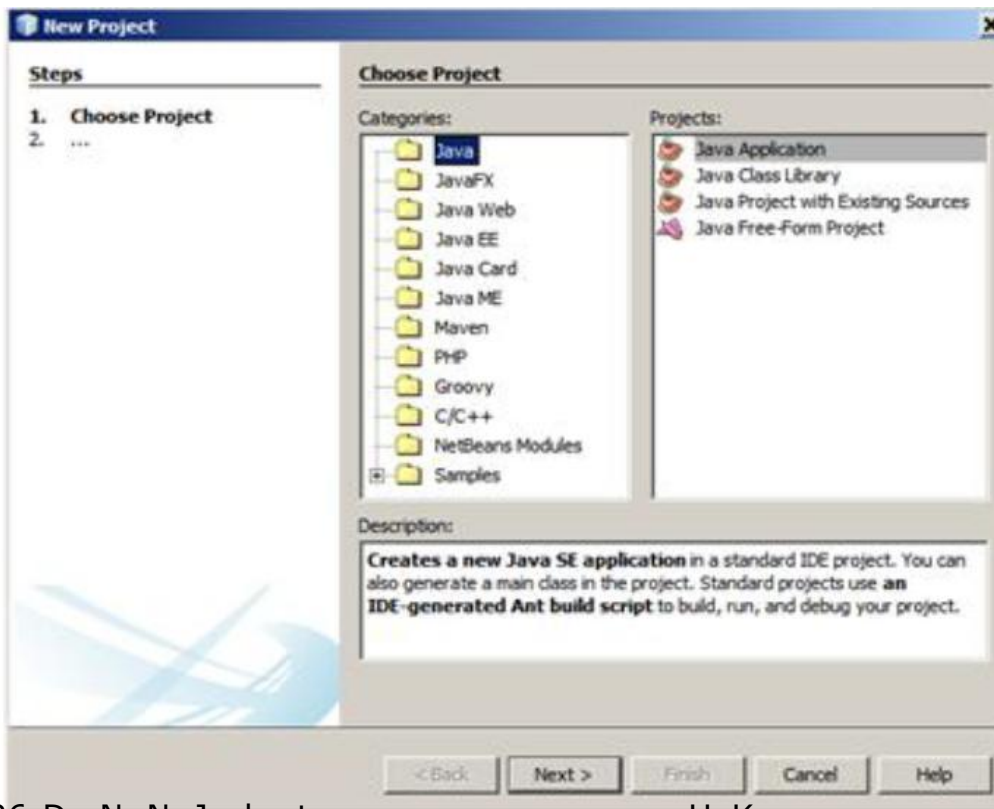
1. Start NetBeans IDE.
2. In the IDE, choose **File > New Project**, as shown in the figure below.



UNIT 2. JAVA JDBC (Java Database Connectivity)

2.6. Creating Java project in NetBeans

3. In the **New Project wizard**, expand the Java category and select Java Application as shown in the figure below. Then click **Next**.



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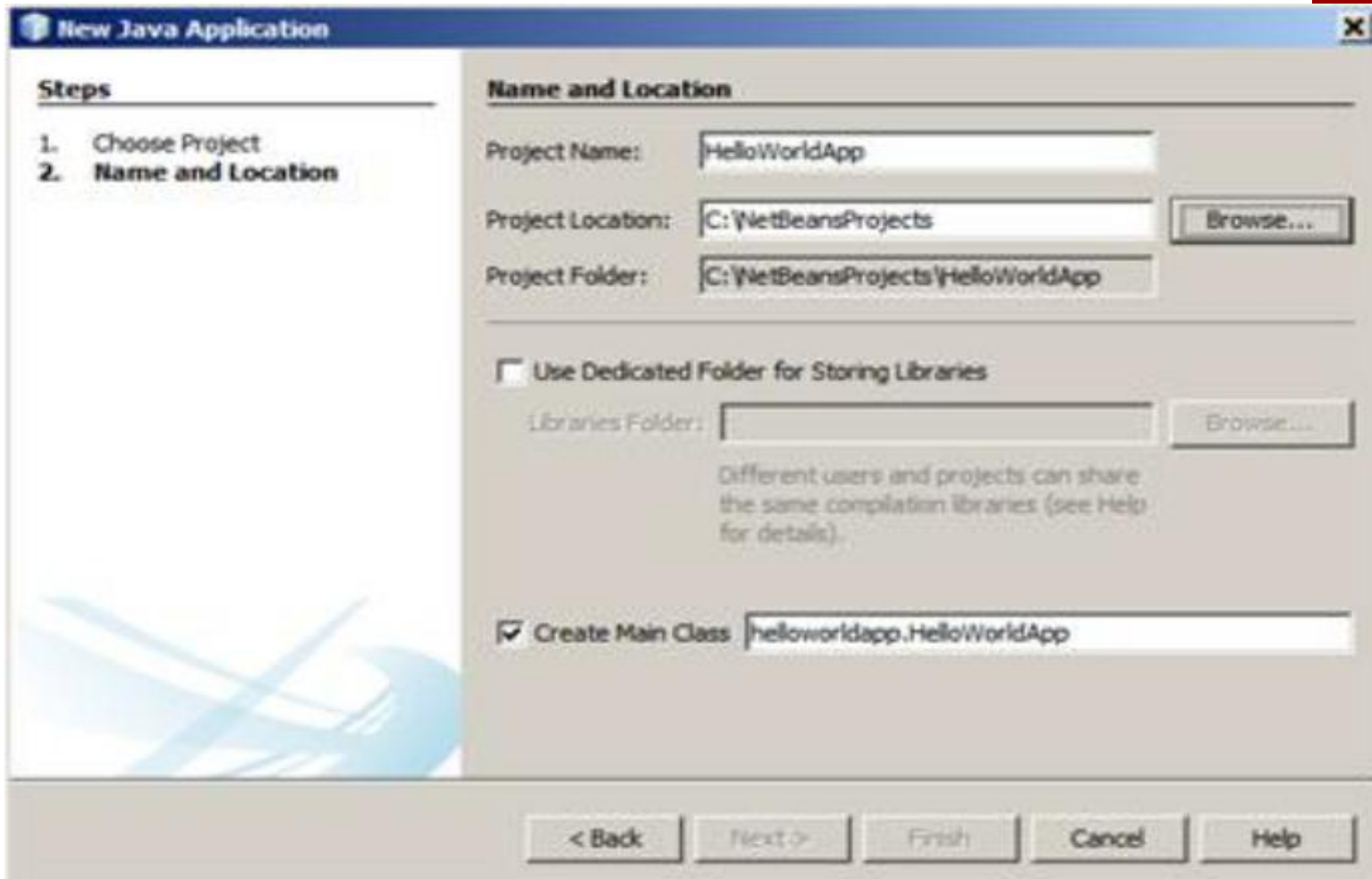
2.6. Creating Java project in NetBeans

In the Name and Location page of the wizard, do the following (as shown in the figure below):

- In the Project Name field, type HelloWorldApp.
- Leave the Use Dedicated Folder for Storing Libraries checkbox unselected.
- In the Create Main Class field, type helloworldapp.HelloWorldApp.

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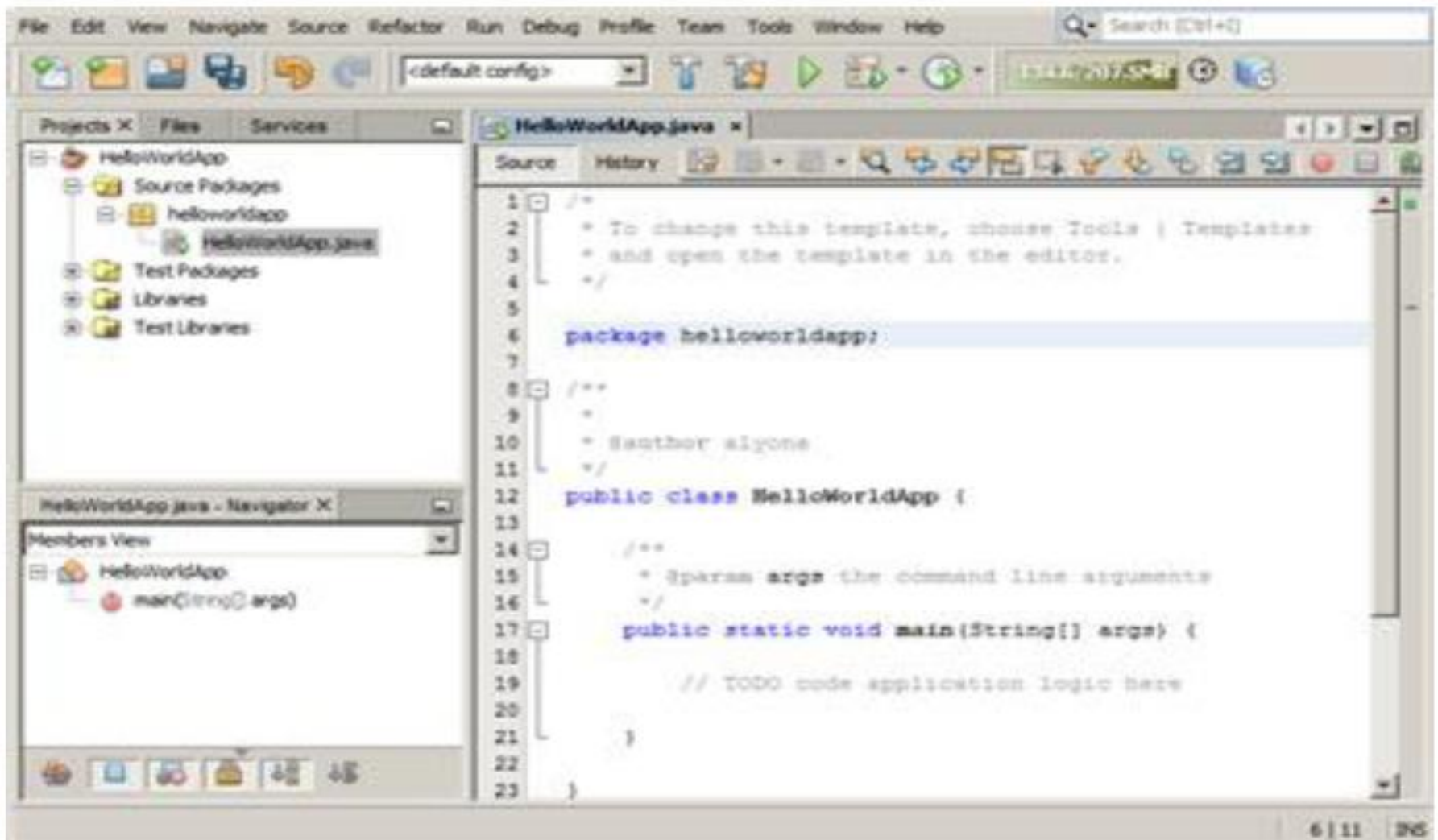
2.6. Creating Java project in NetBeans



Click Finish.

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18.6. Creating Java project in NetBeans



UNIT 2. JAVA JDBC (Java Database Connectivity)

2.7 . Connect to the database

Now write some code to connect to MySQL database. MySQL service is configured on localhost.

So we use Connection and DriverManager Classes so we need to import libraries.

`import java.sql.*;`

```
package testmysql;

import java.sql.*;

/**
 *
 * @author Administrator
 */
public class Main {

    /** Creates a new instance of Main */
    public Main() {

    }

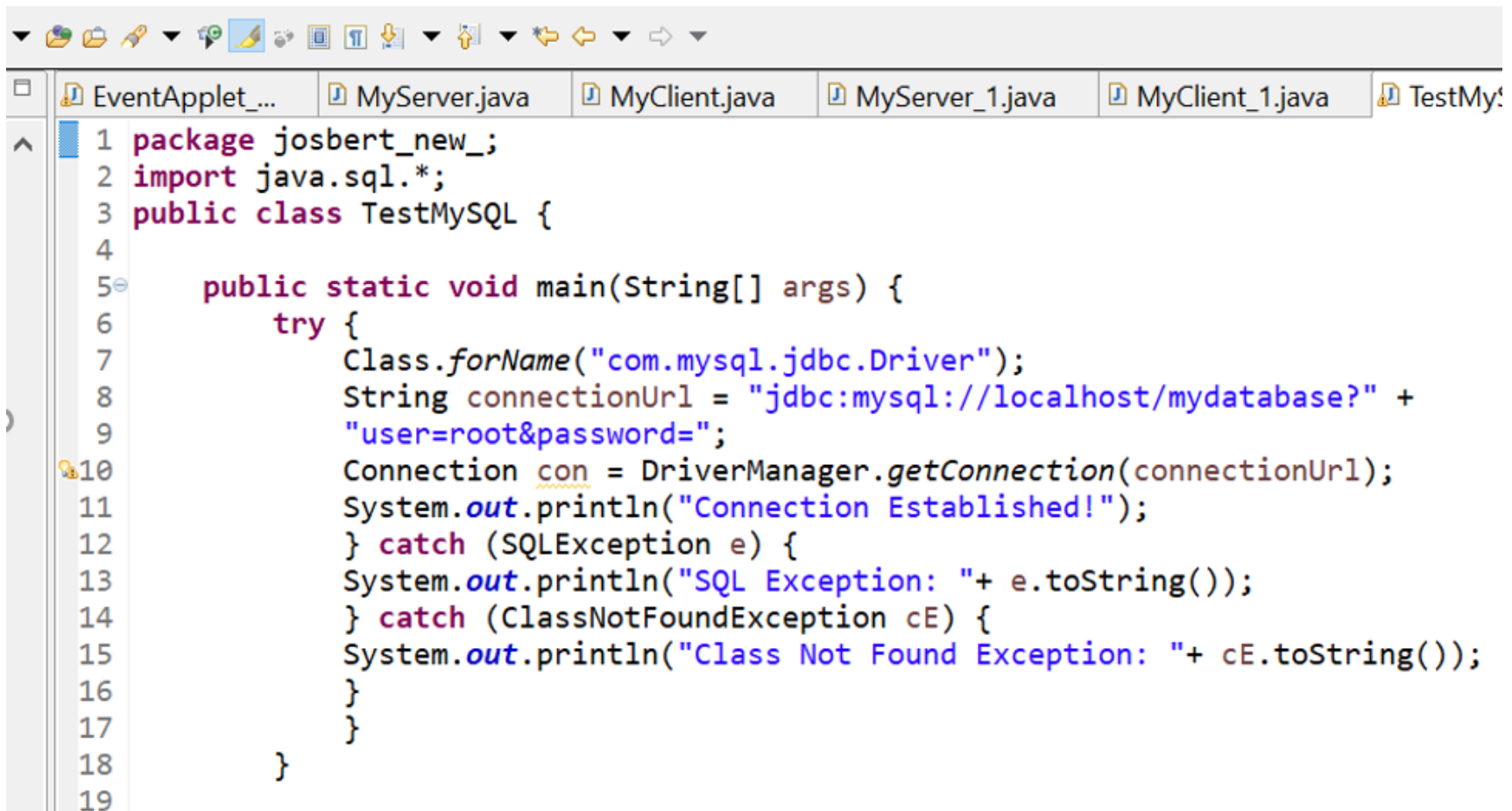
    /**
     * @param args the command line arguments
     */
    public static void main(String[] args) {
        // TODO code application logic here
    }

}
```

UNIT 2. JAVA JDBC (Java Database Connectivity)

2.7 . Connect to the database

The full program to connect to **mydatabase** database:

A screenshot of a Java IDE window. The title bar shows several open files: EventApplet_..., MyServer.java, MyClient.java, MyServer_1.java, MyClient_1.java, and TestMySQL.java. The editor displays the source code for TestMySQL.java. The code is as follows:

```
1 package josbert_new_;
2 import java.sql.*;
3 public class TestMySQL {
4
5     public static void main(String[] args) {
6         try {
7             Class.forName("com.mysql.jdbc.Driver");
8             String connectionUrl = "jdbc:mysql://localhost/mydatabase?" +
9                 "user=root&password=";
10            Connection con = DriverManager.getConnection(connectionUrl);
11            System.out.println("Connection Established!");
12        } catch (SQLException e) {
13            System.out.println("SQL Exception: " + e.toString());
14        } catch (ClassNotFoundException cE) {
15            System.out.println("Class Not Found Exception: " + cE.toString());
16        }
17    }
18 }
19
```


UNIT 2. JAVA JDBC (Java Database Connectivity)

2.7 . Connect to the database

Let's add more comments to this Java program

```
1 package josbert_new_;
2 import java.sql.*; // Importing the JDBC package for database connectivity
3 public class TestMySQL_cmts {
4     // Main method where the program execution starts
5     public static void main(String[] args) {
6         try {
7             // Load the MySQL JDBC driver
8             Class.forName("com.mysql.jdbc.Driver");
9
10            // Define the connection URL for the database
11            // It includes the database location (localhost) and name (mydatabase)
12            // along with the user and password parameters
13            String connectionUrl = "jdbc:mysql://localhost/mydatabase?" +
14                                   "user=root&password=";
15
16            // Establish the connection using DriverManager
17            Connection con = DriverManager.getConnection(connectionUrl);
18
19            // If the connection is established successfully, print a confirmation message
20            System.out.println("Connection Established!");
21        } catch (SQLException e) {
22            // Handle the SQL exceptions that might occur during the database operations
23            System.out.println("SQL Exception: " + e.toString());
24        } catch (ClassNotFoundException cE) {
25            // Handle the exception if the JDBC driver class is not found
26            System.out.println("Class Not Found Exception: " + cE.toString());
27        }
28    }
29 }
```

Fall-2026-Dr. N. N. Josbert

UNIT 2. JAVA JDBC (Java Database Connectivity)

2. 8. Insert data into the database

To insert, update and delete records on SQL Server, you can use the code from retrieve data from database and simply change SQL command and also modify some code a little bit. **On update, we must use `executeUpdate("SQL")` method on statement object instead `executeQuery("SQL")` and the return value will be rows affected instead of a record set.**

```
// SQL insert command
Statement stmt = null;
ResultSet rs = null;
stmt = con.createStatement();
String strSQL = "INSERT INTO students (id,firstname,lastname,email) VALUES
(1,'KABIBI','Jean','kabibi@gmail.com)";
int rowsEffectd = stmt.executeUpdate(strSQL);
System.out.println(rowsEffectd + " rows effected");
```

UNIT 2. JAVA JDBC (Java Database Connectivity)

2. 8. Insert data into the database

```
// SQL insert command
Statement stmt = null;
ResultSet rs = null;
stmt = con.createStatement();
String strSQL = "INSERT INTO students (id,firstname,lastname,email) VALUES
(1,'KABIBI','Jean','kabibi@gmail.com)";
int rowsEffectd = stmt.executeUpdate(strSQL);
System.out.println(rowsEffectd + " rows effected");
```

```
// SQL insert command
Statement stmt = null;
ResultSet rs = null;
stmt = con.createStatement();
String strSQL = "INSERT INTO students (id,firstname,lastname,email) VALUES
(1,'KABIBI','Jean','kabibi@gmail.com)";
int rowsEffectd = stmt.executeUpdate(strSQL);
System.out.println(rowsEffectd + " rows effected"); |
```

UNIT 2. JAVA JDBC (Java Database Connectivity)

2. 8. Insert data into the database

Let's add more comments to this program

```
// Initialize Statement and ResultSet objects
Statement stmt = null;
ResultSet rs = null;

// Create a Statement object using the established database connection
stmt = con.createStatement();

// Define the SQL INSERT command to add a new record to the 'students' table
// The command specifies the target columns (id, firstname, lastname, email)
// and the corresponding values to be inserted
String strSQL = "INSERT INTO students (id, firstname, lastname, email) " + |
                "VALUES (1, 'KABIBI', 'Jean', 'kabibi@gmail.com')";

// Execute the SQL INSERT command using the executeUpdate method
// This method returns the number of rows affected by the operation
int rowsEffected = stmt.executeUpdate(strSQL);

// Print the number of rows that were affected by the INSERT operation
// This tells us how many records were inserted into the database
System.out.println(rowsEffected + " rows effected");
```

UNIT 2. JAVA JDBC (Java Database Connectivity)

2. 8. Insert data into the database

Full Codes for the insert operation:

```

MyClient.java  MyServer_1.java  MyClient_1.java  TestMySQL.java  MysqlCon.java  *TestMySQL_....
1 package josbert_new_;
2 import java.sql.*;
3 public class TestMySQL_ {
4     public static void main(String[] args) {
5         try {
6             Class.forName("com.mysql.jdbc.Driver");
7             String connectionUrl = "jdbc:mysql://localhost/mydatabase?" +
8                 "user=root&password=";
9             Connection con = DriverManager.getConnection(connectionUrl);
10            System.out.println("Connection Established!");
11            Statement stmt = null;
12            ResultSet rs = null; stmt = con.createStatement();
13            String strSQL = "INSERT INTO students (id,firstname,lastname,email) "
14                + "VALUES (1,'KABIBI','Jean','kabibi@gmail.com)";
15            int rowsEffected = stmt.executeUpdate(strSQL);
16            System.out.println(rowsEffected + " rows effected");
17        }
18        catch (SQLException e) {
19            System.out.println("SQL Exception: " + e.toString());
20        } catch (ClassNotFoundException cE) {
21            System.out.println("Class Not Found Exception: " + cE.toString());
22        }
23    }
24 }

```

Fall-2026-Dr. N. N. Josbert UoK

UNIT 2. JAVA JDBC (Java Database Connectivity)

2. 9. UPDATE Command

// SQL update command

```
String strSQL = "UPDATE student SET lastname = 'John'
WHERE firstname = 'KABIBI';
int rowsEffected = stmt.executeUpdate(strSQL);
System.out.println(rowsEffected + " rows effected");
```

```
// SQL update command
String strSQL = "UPDATE student SET lastname = 'John' WHERE firstname =
'KABIBI';
int rowsEffected = stmt.executeUpdate(strSQL);
System.out.println(rowsEffected + " rows effected");
```


UNIT 2. JAVA JDBC (Java Database Connectivity)

2. 9. UPDATE Command

Run Window Help

```

MyServer_1.java  MyClient_1.java  TestMySQL.java  MysqlCon.java  TestMySQL_.java  TestMySQL_2...  » 99+
1  package josbert_new_;
2
3  import java.sql.*;
4  public class TestMySQL_2 {
5
6      /**
7       * @param args the command line arguments
8       */
9      public static void main(String[] args) {
10         try {
11             Class.forName("com.mysql.jdbc.Driver");
12             String connectionUrl = "jdbc:mysql://localhost/mydatabase?" +
13                 "user=root&password=";
14             Connection con = DriverManager.getConnection(connectionUrl);
15             System.out.println("Connection Established!");
16             Statement stmt = null;
17             ResultSet rs = null;
18             stmt = con.createStatement();
19             // SQL update command
20             String strSQL = "UPDATE students SET lastname = 'John' WHERE firstname = 'KABIBI'";
21             int rowsAffected = stmt.executeUpdate(strSQL);
22             System.out.println(rowsAffected + " rows effected");
23         } catch (SQLException e) {
24             System.out.println("SQL Exception: " + e.toString());
25         } catch (ClassNotFoundException cE) {
26             System.out.println("Class Not Found Exception: " + cE.toString());
27         }
28     }
29 }
30 }

```

Output:
1 rows effected

UNIT 2. JAVA JDBC (Java Database Connectivity)

2. 10. DELETE command

// SQL delete command

```
String strSQL = "DELETE FROM students WHERE lastname  
= 'John'";
```

```
int rowsEffected = stmt.executeUpdate(strSQL);
```

```
System.out.println(rowsEffected + " rows effected");
```

```
// SQL delete command
```

```
String strSQL = "DELETE FROM students WHERE lastname = 'John'";
```

```
int rowsEffected = stmt.executeUpdate(strSQL);
```

```
System.out.println(rowsEffected + " rows effected");
```

```
|
```


UNIT 2. JAVA JDBC (Java Database Connectivity)

2. 10. DELETE command

```
1 package josbert_new_;
2
3 import java.sql.*;
4 public class TestMySQL_3 {
5
6     public static void main(String[] args) {
7         try {
8             Class.forName("com.mysql.jdbc.Driver");
9             String connectionUrl = "jdbc:mysql://localhost/mydatabase?" +
10                "user=root&password=";
11             Connection con = DriverManager.getConnection(connectionUrl);
12             System.out.println("Connection Established!");
13             Statement stmt = null;
14             ResultSet rs = null;
15             stmt = con.createStatement();
16             // SQL delete command
17             String strSQL = "DELETE FROM students WHERE lastname = 'John'";
18             int rowsEffected = stmt.executeUpdate(strSQL);
19             System.out.println(rowsEffected + " rows effected");
20         } catch (SQLException e) {
21             System.out.println("SQL Exception: " + e.toString());
22         } catch (ClassNotFoundException cE) {
23             System.out.println("Class Not Found Exception: " + cE.toString());
24         }
25     }
26 }
27
```

Sample Output:
1 rows effected

UNIT 2. GRAPHICAL USER INTERFACE PROGRAMMING

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Exercises:

1. Person Management Data

Person Management

First Name:

Middle Name:

Last Name:

Email:

Phone:

ID	First Name	Middle Name	Last Name	Email	Phone
1	Josbert	NTEZIRIZA	NKERABAHIZI	josbert@gmail.com	0794322298
2	Isaline	UMUKUNZI	UWASE	isaline@gmail.com	09434343434
3	Solange	UMWALI	MWIZA	solange@gmail.com	0743434343
4	Musabe	John	Eric	Eric@gmail.com	4343434343434
5	Hubert	Rugwiro	musabe	hubert@gmail.com	434343434343

SELECT * FROM `person`

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows:

25

 | Filter rows:

Search this table

 | Sort by key:

None

Extra options

←T→

☐ [Edit](#) [Copy](#) [Delete](#)

personId

firstName

middleName

lastName

email

phone

☐ [Edit](#) [Copy](#) [Delete](#)

1

Josbert

NTEZIRIZA

NKERABAHIZI

josbert@gmail.com

0794322298

☐ [Edit](#) [Copy](#) [Delete](#)

2

Isaline

UMUKUNZI

UWASE

isaline@gmail.com

09434343434

☐ [Edit](#) [Copy](#) [Delete](#)

3

Solange

UMWALI

MWIZA

solange@gmail.com

0743434343

☐ [Edit](#) [Copy](#) [Delete](#)

4

Musabe

John

Eric

Eric@gmail.com

4343434343434

☐ [Edit](#) [Copy](#) [Delete](#)

5

Hubert

Rugwiro

musabe

hubert@gmail.com

434343434343

☐ Check all

With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

☐ Show all | Number of rows:

25

 | Filter rows:

Search this table

 | Sort by key:

None

New

Insert

Update

Delete

First

Previous

Next

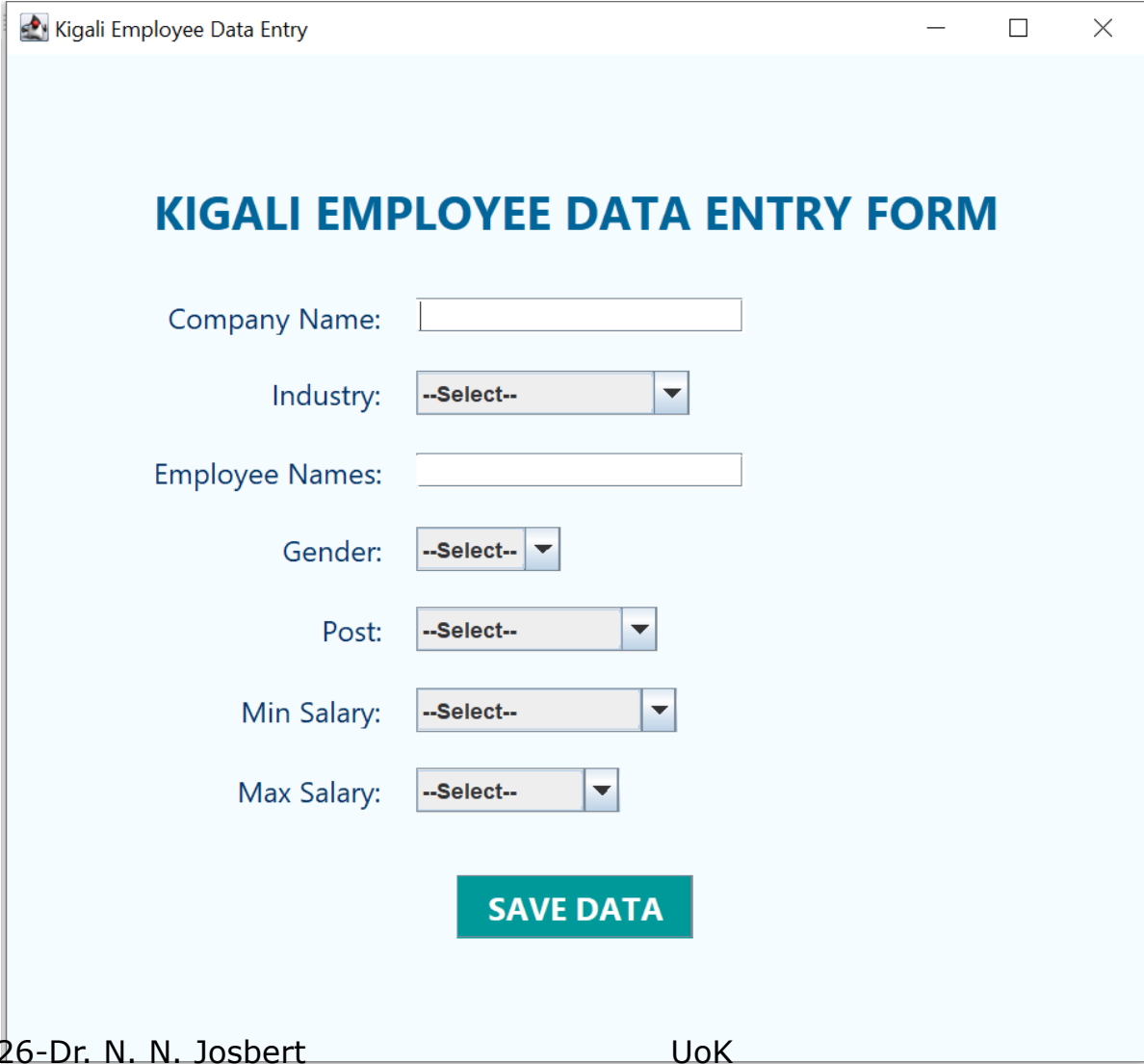
Last

UNIT 2. GRAPHICAL USER INTERFACE PROGRAMMING

Exercises:

2. Kigali Employee Data Entry Form

59



The screenshot shows a window titled "Kigali Employee Data Entry" with a light blue background. The title bar includes a standard Windows icon and window control buttons (minimize, maximize, close). The form content is centered and features the title "KIGALI EMPLOYEE DATA ENTRY FORM" in bold blue text. Below the title, there are seven input fields arranged vertically: "Company Name" (a text box), "Industry" (a dropdown menu with "--Select--"), "Employee Names" (a text box), "Gender" (a dropdown menu with "--Select--"), "Post" (a dropdown menu with "--Select--"), "Min Salary" (a dropdown menu with "--Select--"), and "Max Salary" (a dropdown menu with "--Select--"). At the bottom center of the form is a teal button with the text "SAVE DATA" in white capital letters.

UNIT 2. GRAPHICAL USER INTERFACE PROGRAMMING

Exercises:

2. Kigali Employee Data Entry Form

60

Kigali Employee Data Entry

KIGALI EMPLOYEE DATA ENTRY FORM

Company Name:

Industry:

--Select--

Employee Names:

Gender:

--Select--

SELECT * FROM `kigaliemployeesdata`

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows:

25

 Filter rows: Search this table Sort by key:

None

Extra options

				id	company_name	industry	employee_name	employee_gender	post	min_salary	max_salary	registration_date
☐	Edit	Copy	Delete	1	Tech Solutions	IT	John Doe	Male	Manager	less than 300000 rwf	above 500000	2025-04-25 00:38:33
☐	Edit	Copy	Delete	2	Green Farms	Farming	Jane Smith	Female	CEO	less than 500000 rwf	above 1000000	2025-04-25 00:38:33
☐	Edit	Copy	Delete	3	Jos_Company	Business	Big Jos	Male	Manager	less than 400000 rwf	above 1000000	2025-04-25 00:55:16
☐	Edit	Copy	Delete	4	Vivo	IT	JOSBERTR	Male	CEO	less than 100000 rwf	above 200000	2025-04-27 17:28:11

☐ Check all With selected: [Edit](#) [Copy](#) [Delete](#) [Export](#)

☐ Show all | Number of rows:

25

 Filter rows: Search this table Sort by key:

None

UNIT 2. GRAPHICAL USER INTERFACE PROGRAMMING

Exercises:

3. Employee Record System

Menu | Employee Record System

— □ ×

MENU

 Add Employee

 Increase Salary

 Salary Information

 View Employees Data

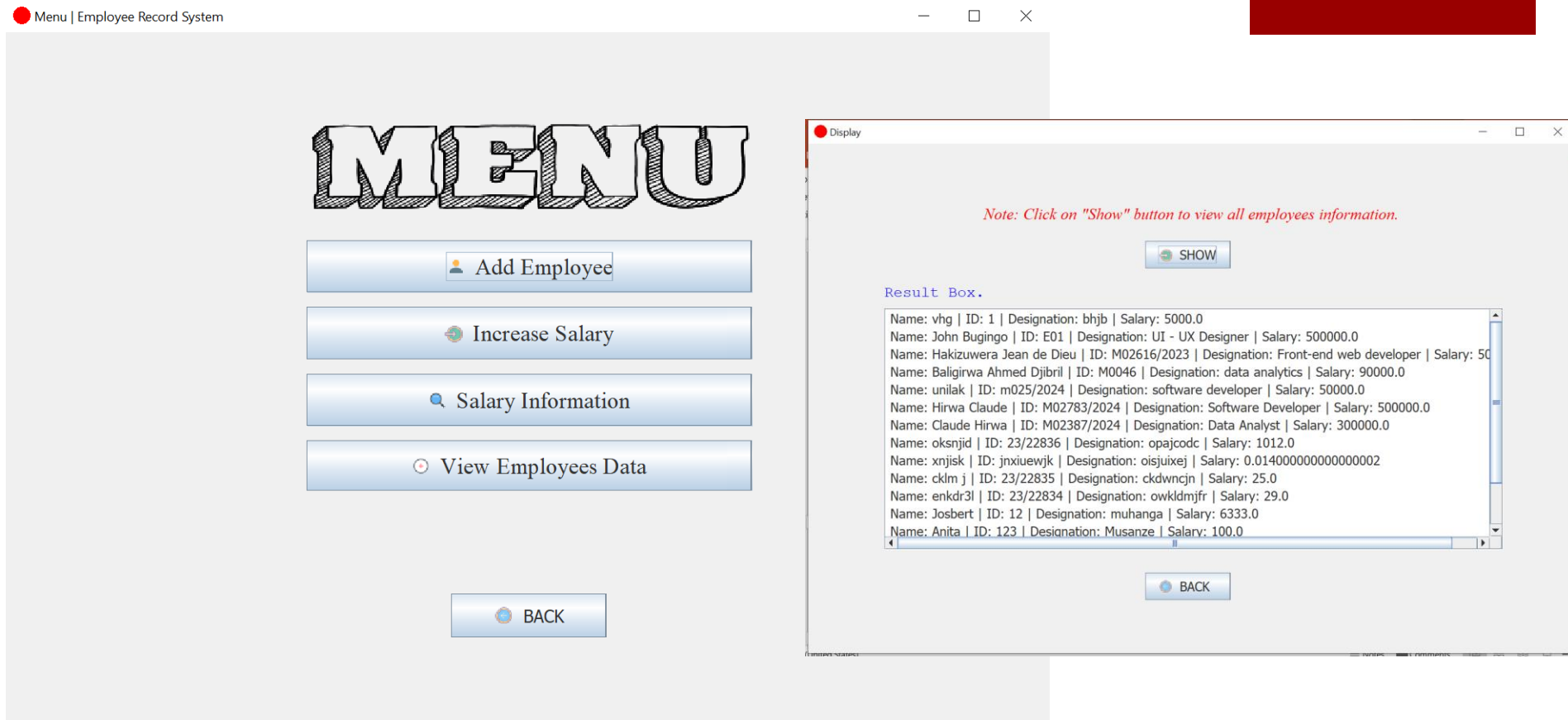
 BACK

UNIT 2. GRAPHICAL USER INTERFACE PROGRAMMING

Exercises:

3. Employee Record System

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UNIT 2. GRAPHICAL USER INTERFACE PROGRAMMING

Exercises:

4. User Management System

63

User Management System

Search User

Search (ID/Name):

User Details

Full Name:

Email:

Phone:

Country:

Actions

ID	FULL NAME	EMAIL	PHONE	COUNTRY
1	John Doe	john.doe@example.com	1234567890	United States
2	Jane Smith	jane.smith@example.com	0987654321	Canada
3	Robert Johnson	robert.j@example.com	5551234567	United Kingdom
7	Josbert NKERABAHIZI	Josbertnn@gmail.com	084343533	France

CHAPTER 17 GRAPHICAL USER INTERFACE PROGRAMMING

Exercises:

64

4. User Management System

User Management System

Search User

Search (ID/Name):

Search

SELECT * FROM `users`

☐ Profiling

[\[Edit inline \]](#)

[\[Edit \]](#)

[\[Explain SQL \]](#)

[\[Create PHP code \]](#)

[\[Refresh \]](#)

☐ Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

Extra options

↔

⌵

				id	fullname	email	phone	country	registration_date
☐	Edit	Copy	Delete	1	John Doe	john.doe@example.com	1234567890	United States	2025-04-25 00:41:17
☐	Edit	Copy	Delete	2	Jane Smith	jane.smith@example.com	0987654321	Canada	2025-04-25 00:41:17
☐	Edit	Copy	Delete	3	Robert Johnson	robert.j@example.com	5551234567	United Kingdom	2025-04-25 00:41:17
☐	Edit	Copy	Delete	7	Josbert NKERABAHIZI	Josbertnn@gmail.com	084343533	France	2025-04-25 01:12:29

⬆

☐ Check all

With selected:

[Edit](#)

[Copy](#)

[Delete](#)

[Export](#)

☐ Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

UNIT 2. GRAPHICAL USER INTERFACE PROGRAMMING

Exercises:

5. Car Rent System

65

LOSRENT LOGIN PAGE

Login Form

Use valid Username and password to login

Username

admin

Password

Login

Exit

UNIT 2. GRAPHICAL USER INTERFACE PROGRAMMING

Exercises:

5. Car Rent System

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ADMIN PANEL

CAR REGISTRATION

Manage Customer

RENTAL

Return a Car

LOGOUT

UNIT 2. GRAPHICAL USER INTERFACE PROGRAMMING

Exercises:

5. Car Rent System

67

Car_RegNo	Make	Model	Plate No	Available
CA001	Toyota	Corolla	RAA123A	No
CA002	Honda	Civic	RAB456B	Yes
CA003	Mazda	CX-5	RAC789C	Yes

UNIT 2. GRAPHICAL USER INTERFACE PROGRAMMING

Exercises:

5. Car Rent System

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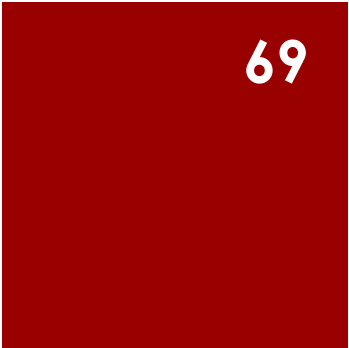
The screenshot displays a graphical user interface for a Car Rent System. On the left, there is a 'Customer Form' titled 'Customer Registration Form'. It contains five input fields: 'Customer ID' (with 'C0004' entered), 'Customer Names', 'Address' (a text area), 'Phone No', and 'Password'. Below these fields are five buttons: 'Add', 'Delete', 'Update', 'Back', and 'Refresh'. On the right, there is a table with four columns: 'Customer ID', 'Names', 'Address', and 'Phone No'. The table contains three rows of data.

Customer ID	Names	Address	Phone No
C0001	John Doe	123 Main St, Kigali	0781234567
C0002	fdfdf	fdfdf	0788843434
C0003	Josbert	Rwanda, Musanze	0794322298

UNIT 2. GRAPHICAL USER INTERFACE PROGRAMMING

Exercises:

5. Car Rent System



Customer Form

Customer Registration Form

Customer ID

C0004

Customer Names

Address

Phone No

Password

Add

Delete

Update

Back

Refresh

Customer ID	Names	Address	Phone No
C0001	John Doe	123 Main St, Kigali	0781234567
C0002	fefdf	fefdf	0788843434
C0003	Josbert	Rwanda, Musanze	0794322298

Filters

Containing the word:

	Table	Action	Rows	Type	Collation	Size	Overhead
☐	car	Browse Structure Search Insert Empty Drop	3	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐	customer	Browse Structure Search Insert Empty Drop	3	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐	rent	Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐	returncar	Browse Structure Search Insert Empty Drop	0	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐	user	Browse Structure Search Insert Empty Drop	3	InnoDB	utf8mb4_general_ci	32.0 KiB	-
	5 tables	Sum	10	InnoDB	utf8mb4_general_ci	160.0 KiB	0 B

☐ Check all

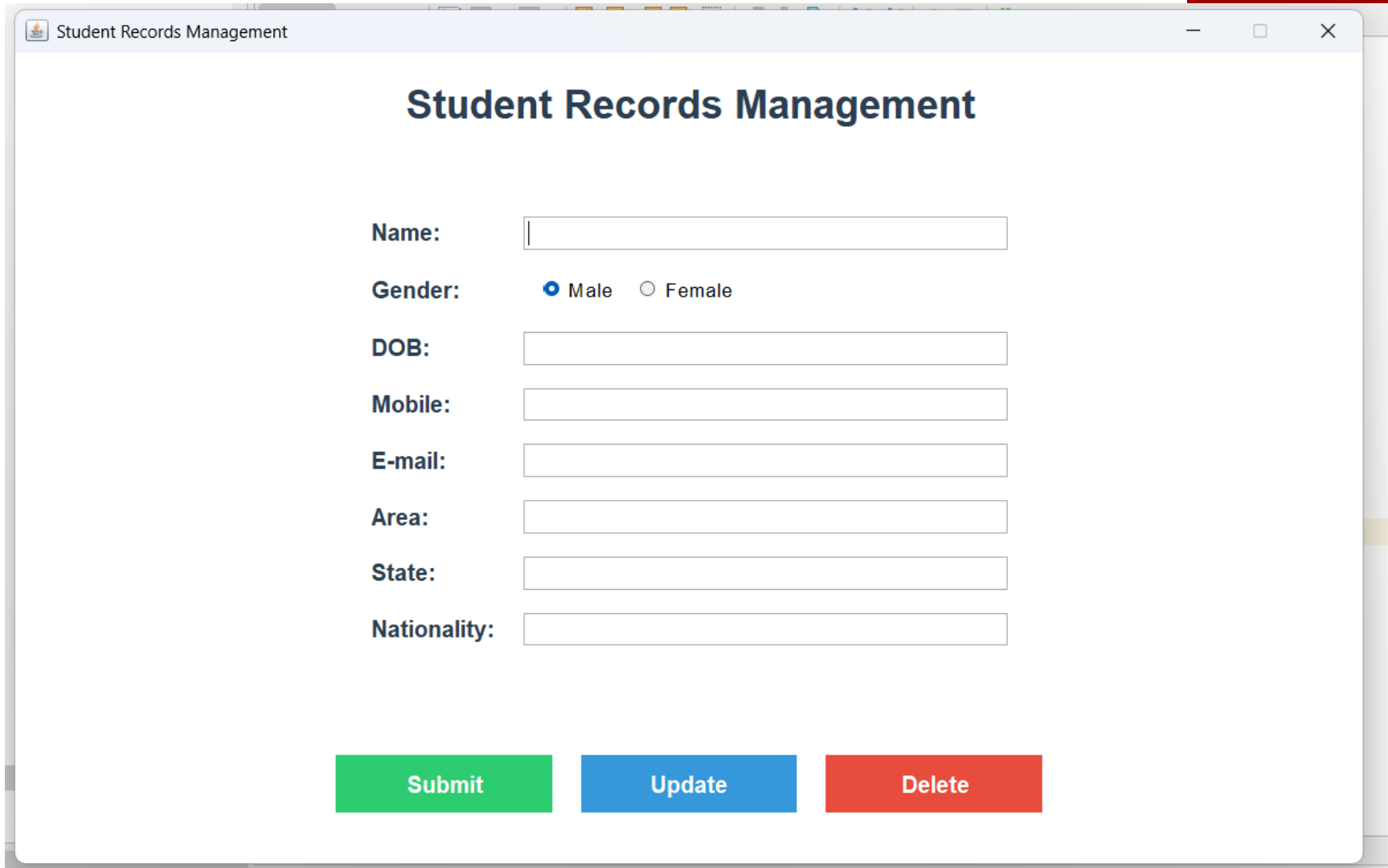
With selected:

UNIT 2. GRAPHICAL USER INTERFACE PROGRAMMING

Exercises:

6. Student Record Management

70



The screenshot shows a web application window titled "Student Records Management". The window contains a form with the following fields and controls:

- Name:** A text input field.
- Gender:** Radio buttons for "Male" (selected) and "Female".
- DOB:** A text input field.
- Mobile:** A text input field.
- E-mail:** A text input field.
- Area:** A text input field.
- State:** A text input field.
- Nationality:** A text input field.

At the bottom of the form, there are three buttons: "Submit" (green), "Update" (blue), and "Delete" (red).

UNIT 2. GRAPHICAL USER INTERFACE PROGRAMMING

Exercises:

6. Student Record Management

71

The screenshot shows a web application window titled "Student Records Management". Inside the window, there is a form with the following fields and values:

- Name:** NKERABAHIZI Josbert
- Gender:** ☒ Male ☐ Female
- DOB:** 1920-05-06
- Mobile:** 0788547461
- E-mail:** josberr@gmail.com
- Area:** Kigali
- State:** Kigali
- Nationality:** Rwandan

At the bottom of the form, there are three buttons: "Submit" (green), "Update" (blue), and "Delete" (red). A modal dialog box titled "Success" is overlaid on the right side of the form, displaying an information icon and the message "Student record inserted successfully!". The dialog has an "OK" button.

UNIT 2. GRAPHICAL USER INTERFACE PROGRAMMING

Exercises:

6. Student Record Management

72

☐ Show all | Number of rows: 25 ▾ | Filter rows:

Sort by key:

None ▾

+ Options

				id	name	gender	dob	mobile	email	area	state	nationality	created_at	updated_at			
☐		Edit		Copy		Delete	1	John Doe	Male	2000-01-15	9876543210	john@example.com	Downtown	California	USA	2025-11-09 21:00:03	2025-11-09 21:00:03
☐		Edit		Copy		Delete	2	Josbert	Male	1987-12-09	1233444222	josbert1@gmail.com	kigali	RWANDA	RWANDA	2025-11-09 21:06:39	2025-11-09 21:06:39
☐		Edit		Copy		Delete	3	Isaline	Male	2000-12-12	4444444444	isaline@gmail.com	france	france	france	2025-11-09 21:13:17	2025-11-09 21:14:48
☐		Edit		Copy		Delete	4	NKERABAHIZI Josbert	Male	1920-05-06	0788547461	josberr@gmail.com	Kigali	Kigali	Rwandan	2026-04-07 15:01:30	2026-04-07 15:01:30

☐ Check all | With selected: Edit Copy Delete Export

☐ Show all | Number of rows: 25 ▾ | Filter rows:

UNIT 2. GRAPHICAL USER INTERFACE PROGRAMMING

Exercises:

7. UoK Student Registration System

73

University of Kigali - Student Registration

U

UNIVERSITY OF KIGALI
STUDENT REGISTRATION SYSTEM · MYSQL DATABASE

Student Registration Form
Fields marked * are mandatory · Data stored in MySQL

Full Name *

N. N. Josbert

Gender *

☒ Male ☐ Female

Password *

.....

Confirm Password *

.....

Date of Birth *

3

▼

February

2023

▼

Mobile Number *

0788547461

E-mail Address *

josbert@gmail.com

Area *

Nyarugenge

▼

State / Province *

Kigali City

▼

Nationality *

Rwandan

▼

Clear

Submit

Fall-2026-Dr. N. N. JosbertUniversity of Kigali · Student Registration System · Java Spring + MySQL (JDBC)UoK

UNIT 2. GRAPHICAL USER INTERFACE PROGRAMMING

Exercises:

7. UoK Student Registration System

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The screenshot displays a Java Swing window titled "University of Kigali - Student Registration". The window has a purple header bar with the university logo and name. Below the header, there's a "Student Registration Form" section with a note: "Fields marked * are mandatory - Data stored in MySQL". The form contains several input fields: "Full Name *" (filled with "N. N. Josbert"), "Gender *" (radio buttons for "Male" and "Female", with "Male" selected), "Password *" (masked with "....."), "Confirm Password *" (empty), "Date of Birth *" (dropdown for "3" and "February"), "Mobile Number *" (filled with "0788547461"), "Email" (filled with "josbert@gmail.com"), "Area *" (dropdown for "Nyarugenge"), "State / Province *" (dropdown for "Kigali City"), and "Nationality *" (dropdown for "Rwandan"). A "Success" dialog box is overlaid on the form, displaying an information icon, the text "Student registered successfully! Welcome, N. N. Josbert!", and an "OK" button. At the bottom right of the form are "Clear" and "Submit" buttons. The footer of the window reads "University of Kigali - Student Registration System - Java Swing + MySQL (JDBC)".

UNIT 2. GRAPHICAL USER INTERFACE PROGRAMMING

Exercises:

7. UoK Student Registration System

75

University of Kigali - Student Registration

U

UNIVERSITY OF KIGALI

STUDENT REGISTRATION SYSTEM · MYSQL DATABASE

Server: 127.0.0.1 » Database: uk_form_database » Table: students

BrowseStructureSQLSearchInsertExportImportPrivilegesOperationsTrackingTriggers

Showing rows 0 - 3 (4 total, Query took 0.0007 seconds.)

SELECT * FROM `students`

[Edit inline][Edit][Explain SQL][Create PHP code][Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Sort by key: None

+ Options

☐ Edit Copy Delete

☐ Edit Copy Delete

☐ Edit Copy Delete

☐ Edit Copy Delete

student_id	full_name	gender	password	date_of_birth	mobile_number	email_address	area	province	nationality	registered_at
1	NTEZIRIZA Josbert	Male	josbert12	2024-01-02	0788547461	josbert@gmail.com	Nyarugenge	Kigali City	Rwandan	2026-03-15 18:15:02
2	UWIMANA Irarera Alice	Female	uwimana12	2022-02-02	1234567890	uwimana@gmail.com	Nyarugenge	Kigali City	Ugandan	2026-03-15 18:25:37
3	Mugabo Zidane	Male	hubert12	2023-02-02	0788887878	hubert@gmail.com	Nyarugenge	Kigali City	Rwandan	2026-03-17 14:40:54
4	N. N. Josbert	Male	kigali	2023-02-03	0788547461	josbert@gmail.com	Nyarugenge	Kigali City	Rwandan	2026-04-07 15:12:06

☐ Check all | With selected: Edit Copy Delete Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

Print viewExportDisplay chartCreate view

ClearSubmit

Fall-2026-Dr. N. N. JosbertUniversity of Kigali · Student Registration System · Java Swing + MySQL (JDBC)

UNIT 2. GRAPHICAL USER INTERFACE PROGRAMMING

Exercises:

8. TUZA Hospital Patient Management System

76

Welcome to TUZA HOSPITAL Patient Management System

Patient Information

Patient ID:

First Name:

Last Name:

Address:

Add View All Edit Delete Clear

Patient ID	First Name	Last Name	Address
3	John	mugabo	Kigali
3232	N. N.	Josbert	Kigali - Musanze
1341414	Dr Musabe	Jean Bosco	Kigali
5454545	rttrtrtrtr	trtrtrt	trtrtrtr

UNIT 2. GRAPHICAL USER INTERFACE PROGRAMMING

Exercises:

8. TUZA Hospital Patient Management System

77

 TUZA HOSPITAL Patient Management System

Welcome to TUZA HOSPITAL Patient Management System

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Sort by key: None

+ Options

		PatientID	FirstName	LastName	Address
☐	Edit Copy Delete	3	John	mugabo	Kigali
☐	Edit Copy Delete	3232	N. N.	Josbert	Kigali - Musanze
☐	Edit Copy Delete	1341414	Dr Musabe	Jean Bosco	Kigali
☐	Edit Copy Delete	5454545	rttrtrtrtr	trtrtrtr	trtrtrtr

☐ Check all | With selected: Edit Copy Delete Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table

Query results operations

 Print view  Export  Display chart  Create view

UNIT 2. GRAPHICAL USER INTERFACE PROGRAMMING

Exercises:

9. Calculator



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